Introduction to Biostatistics

208141

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2026-10-11

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Preface

This book is an introduction to biostatistics for first-year health science students. It covers data types, summarization, probability distributions, estimation, hypothesis testing, and an introduction to statistical modeling.

All examples use health and clinical contexts.

Data in Health Science

Introduction

Why Does This Matter?

Every day, health professionals make decisions based on data — patient vital signs, lab results, medication dosages, and clinical outcomes. Understanding **what data is** and **how to classify it correctly** is the very first step toward making safe, evidence-based decisions in clinical practice.
In this chapter, we lay the foundation for everything that follows in this course.

Data Types

What is Data?

Data are pieces of information collected from observations or measurements. In health science, data come from patients, experiments, surveys, or medical records.

> **Example:** A health worker records the blood pressure, age, gender, pain score, and discharge status of every patient on the ward. Each of these recorded pieces of information is *data*.

Data can be broadly divided into two major types:

- **Categorical Data** — data that represent categories or groups
- **Numerical Data** — data that represent measurable quantities

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Categorical Data

Categorical data place individuals into groups. There are two subtypes:

Nominal Data

Nominal data have **no natural order** among categories. You can only say things are *different*, not *bigger* or *smaller*.

Nominal Examples

- Blood type: A, B, AB, O
- Gender: Male, Female
- Disease diagnosis: Diabetes, Hypertension, Asthma
- Ward type: Medical, Surgical, Pediatric, ICU

Ordinal Data

Ordinal data have **a natural order**, but the *distance* between categories is not necessarily equal.

Ordinal Examples

- Pain score (0–10 numeric rating scale): 0 = no pain, 10 = worst pain imaginable
- Patient satisfaction: Very dissatisfied < Dissatisfied < Neutral < Satisfied < Very satisfied
- Level of consciousness (AVPU): Alert > responds to Voice > responds to Pain > Unresponsive
- Education level: Primary < Secondary < Bachelor < Graduate

Pain Scores Are Ordinal in This Course

A pain score is written as a number (0–10), but the numbers are only **ranks**. A change from 2 to 3 does not necessarily mean the same amount of extra pain as a change from 8 to 9, and a score of 6 is not "twice as painful" as a score of 3.
In this course we always treat pain scores as **ordinal data**. This decides how we summarize them (median, Chapter 2) and which test we use to compare them (rank tests, Chapter 5).

Numerical Data

Numerical data represent actual measured or counted quantities. There are two subtypes:

Discrete Data

Discrete data are **counted** and can only take **whole number** values. There are no meaningful values between two consecutive counts.

Discrete Examples

- Number of patients admitted per day: 0, 1, 2, 3, …
- Number of medication errors per month
- Number of children a patient has
- Number of hospital readmissions

Continuous Data

Continuous data are **measured** and can take **any value** within a range, including decimals.

Continuous Examples

- Body temperature: 36.5°C, 37.2°C, 38.8°C
- Blood pressure: 120/80 mmHg, 135/85 mmHg
- Body weight: 58.3 kg, 72.5 kg
- Blood glucose level: 95.4 mg/dL

Summary: Data Type Classification

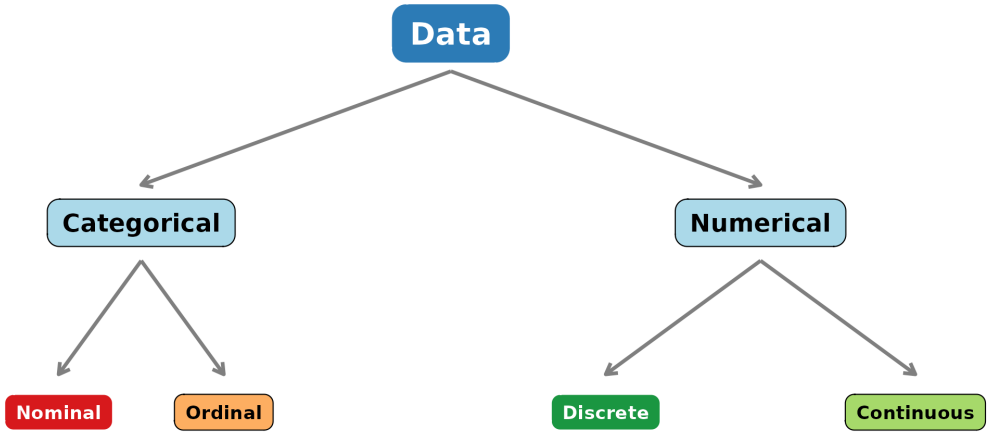


Figure 1: Classification of Data Types in Health Science

Example 1: Classify Each Variable

A researcher collects the following information from 50 patients admitted to a medical ward:

Variable	Recorded Values
Patient ID	001, 002, 003, …
Age	45, 62, 38, … (years)
Gender	Male, Female
Blood type	A, B, AB, O
Systolic blood pressure	118, 135, 142, … (mmHg)
Pain score	0–10 scale
Number of medications	1, 2, 3, …
Discharge status	Recovered, Transferred, Deceased

Solution

Variable	Data Type	Reason
Patient ID	— (Identifier, not analyzed)	Just a label
Age	**Numerical – Continuous**	Measured, can be 45.5 years
Gender	**Categorical – Nominal**	Two categories, no order
Blood type	**Categorical – Nominal**	Four categories, no order
Systolic BP	**Numerical – Continuous**	Measured in mmHg, any value
Pain score	**Categorical – Ordinal**	Has order ($0 < 1 < \ldots < 10$), but gaps unequal
No. of medications	**Numerical – Discrete**	Counted, whole numbers only
Discharge status	**Categorical – Nominal**	Three categories, no natural order

Population and Sample

Population

A **population** is the **complete set** of all individuals or observations that we are interested in studying.

In health research, populations are usually very large and it is **impractical or impossible** to collect data from everyone.

> **Example:** All patients with Type 2 diabetes in Thailand.

Sample

A **sample** is a **subset** of the population that is actually studied. We use the sample to draw conclusions (*make inferences*) about the whole population.

> **Example:** 200 patients with Type 2 diabetes selected from 5 hospitals in Bangkok.

Key Principle

A good sample must be **representative** of the population — meaning it should reflect the characteristics of the whole group, not just one part of it.

Key Terms

Table 1: Important Statistical Terms

Term	Definition	Health Science Example
Population	The entire group of interest	All health-care workers in Thailand
Sample	A selected subset of the population	500 health-care workers randomly selected from 10 hospitals

Parameter	A numerical summary describing the population (e.g., population mean mu)	The true average age of ALL health-care workers in Thailand
Statistic	A numerical summary calculated from the sample (e.g., sample mean xbar)	The average age calculated from our 500 workers
Variable	A characteristic being measured or observed (e.g., age, blood pressure)	Age, gender, profession, years of experience
Observation	A single recorded value for one individual	Worker A: age = 28, female, 3 years experience

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Population vs. Sample: Visual Concept

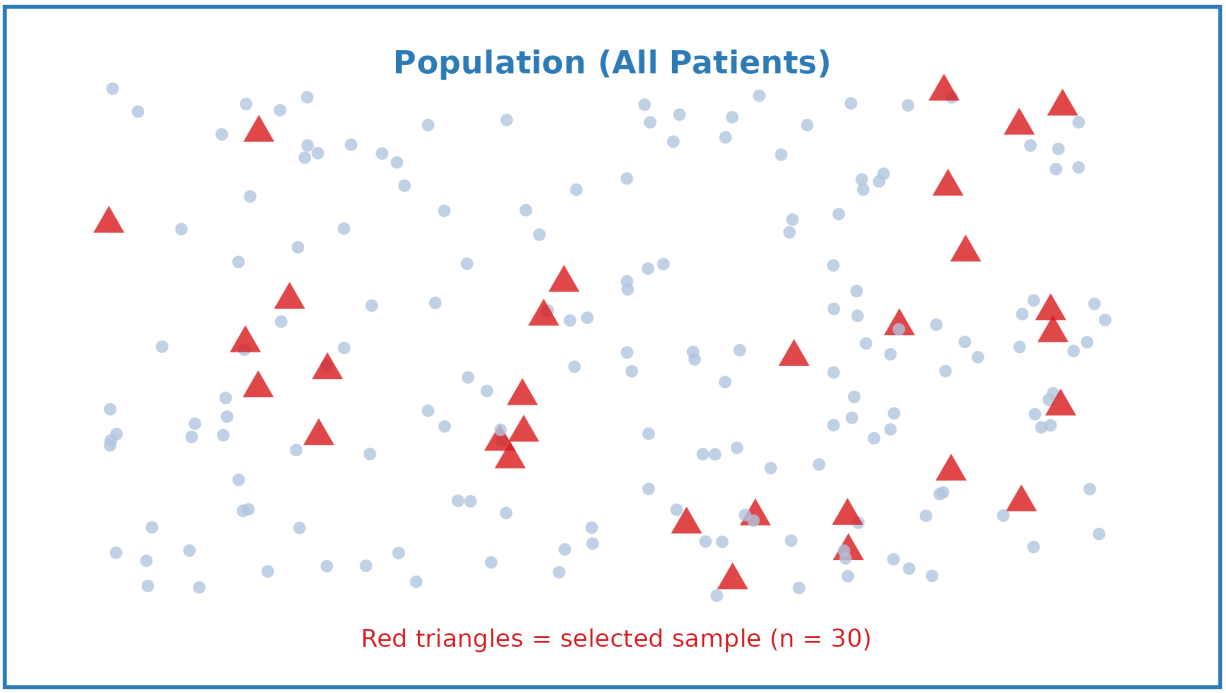


Figure 1: The sample is drawn from the population and used to make inferences

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Example 2: Population or Sample?

For each situation below, identify whether the group described is a **population** or a **sample**:

(a) A hospital wants to know the average job satisfaction of all 320 staff working there. The HR department surveys all 320 staff.

(b) A researcher wants to study pain levels in post-operative patients across Thailand. She recruits 150 patients from 3 hospitals in Chiang Mai.

(c) A student on clinical placement studies the blood glucose levels of the 8 diabetic patients currently assigned to her ward.

Solution

(a) This is the **Population** — all 320 staff are surveyed; no one is left out. The result is a *parameter*.
(b) This is a **Sample** — the 150 patients are only a *subset* of the group of interest (all post-operative patients in Thailand). The result is a *statistic* used to estimate the population parameter. *(Whether this sample is* representative *is a separate question: patients from one province may not reflect the whole country.)*
(c) It depends on **who the student wants to draw conclusions about**:

- If she wants to describe only these 8 patients, they are the whole group of interest, so they are a **Population**.
- If she wants to draw conclusions about diabetic patients in general, the 8 patients are a small **Sample**.

The same group can be a population in one study and a sample in another — it is the *research question* that decides.

Key Takeaways

Chapter 1 Summary

1. **Data** are recorded pieces of information collected from patients or research subjects.
2. Data are classified into two main types:
   - **Categorical**: Nominal (no order) and Ordinal (has order)
   - **Numerical**: Discrete (counted, whole numbers) and Continuous (measured, any value)
3. Identifying the **correct data type** is critical — it determines which statistical methods are appropriate.
4. A **population** is the entire group of interest; a **sample** is the subset we actually study.
5. We use **statistics** calculated from samples to estimate **parameters** of the population — this is the core idea behind all of biostatistics.

Exercises

Exercise 1

A public health officer collects the following data from patients visiting a community health clinic:

- Weight (kg)
- Presence of hypertension (Yes / No)
- Severity of hypertension (Mild / Moderate / Severe)
- Number of clinic visits in the past year
- Ethnicity
- Body temperature (°C)
- Diastolic blood pressure (mmHg)

Classify each variable into the correct data type and explain your reasoning.

Exercise 2

A researcher is studying the prevalence of depression among elderly patients in long-term care facilities across the country. She selects 200 patients from 4 facilities in the central region.
(a) What is the population in this study?
(b) What is the sample?
(c) If the average depression score of the 200 patients is 14.2, is this number a *parameter* or a *statistic*? Explain.

Exercise 3

Look at the following list and decide whether each is an example of a **discrete** or **continuous** variable:

1. The number of patients who fell during a hospital shift
2. The time (in minutes) a patient waited before being seen by a doctor
3. The number of syringes used during a procedure
4. A patient's oxygen saturation level (SpO %)
5. The number of newborns delivered in a hospital per week

End of Chapter 1

Coming Up Next

Chapter 2: Data Summarization — How do we make sense of a large set of patient data? We will learn to calculate means, medians, standard deviations, and draw charts that tell a clear story.

Data Summarization

Introduction

Why Does This Matter?

Imagine you have blood pressure readings from 80 patients on your ward. Looking at 80 individual numbers tells you very little. But if someone says *"the average systolic BP is 138 mmHg and most patients fall between 120 and 155 mmHg"* — now you have a clear picture.

Data summarization is the process of turning raw data into meaningful, easy-to-understand information using numbers and charts. This is one of the most practical skills in clinical practice and health research.

Numerical Summarization

Numerical summarization uses **numbers** to describe the key features of data. There are two important aspects we want to describe:

1. **Central tendency** — Where is the "middle" or "typical" value?
2. **Variability** — How spread out are the values?

Measures of Central Tendency

Mean (Arithmetic Average)

The **mean** is calculated by adding all values and dividing by the number of observations.

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Where:

- $\bar{x}$ = sample mean (read as "x-bar")
- x_i = each individual value
- n = total number of observations
- $\sum$ = "sum of all"

When to Use the Mean

The mean works best for **continuous or discrete numerical data** that is **roughly symmetric** (not heavily skewed) and has **no extreme outliers**. A single extreme value can pull the mean far from the typical value.

Median

The **median** is the **middle value** when all observations are arranged in order.

- If n is **odd**: the median is the value at position $\frac{n+1}{2}$
- If n is **even**: the median is the **average of the two middle values** at positions $\frac{n}{2}$ and $\frac{n}{2}+1$

When to Use the Median

The median is preferred when data is **skewed** or has **outliers**, because it is not affected by extreme values. It is also the appropriate measure for **ordinal data**.

Mode

The **mode** is the value that appears **most frequently** in the data.

- A dataset can have **one mode** (unimodal), **two modes** (bimodal), or **no mode** (all values appear equally).
- The mode is the only measure of central tendency appropriate for **nominal data**.

Choosing the Right Measure

Table 1: Choosing the Right Measure of Central Tendency

Measure	Best For	Example
Mean	Numerical data (continuous or discrete), symmetric distribution	Average patient age, average blood glucose
Median	Numerical data with skewness or outliers; Ordinal data	Average length of hospital stay (skewed by long-stay patients)
Mode	Nominal data; finding the most common category	Most common blood type on a ward, most common reason for admission

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Measures of Variability

Knowing the average alone is not enough. Two wards might have the same average patient age but very different spreads — one ward may have all patients aged 55–65, another may have patients aged 20–90. Variability tells us how **spread out** the data are.

Range

The **range** is the simplest measure of spread:

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

> Limitation
>
> The range only uses two values (max and min) and is very sensitive to outliers.

Variance

The **variance** measures the average squared distance of each value from the mean.

Population variance:

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

Sample variance (used more often in practice):

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

Where:

- μ = population mean, $\bar{x}$ = sample mean
- N = population size, n = sample size
- $n-1$ in the denominator corrects for the fact that we're using a sample

The formula above is the **definition**: it shows that the variance is built from the distances $x_i - \bar{x}$. For calculating by hand, the same value is easier to get from two column totals, $\sum x$ and $\sum x^2$ — no deviations needed, and no rounding of $\bar{x}$ along the way:

Computational Formula for s^2 (Use This by Hand)

$$s^2 = \frac{n\sum x^2 - \left(\sum x\right)^2}{n(n-1)}$$

Note that $\sum x^2$ (square each value, then add) is **not** the same as $\left(\sum x\right)^2$ (add the values, then square the total). The numerator $n\sum x^2 - \left(\sum x\right)^2$ appears again in Chapter 6 (correlation and regression).

Standard Deviation

The **standard deviation (SD)** is the square root of the variance. It is expressed in the **same units as the original data**, making it much easier to interpret than variance.

$$s = \sqrt{s^2} = \sqrt{\frac{n\sum x^2 - \left(\sum x\right)^2}{n(n-1)}}$$

Interpreting Standard Deviation

A **small SD** means values are clustered closely around the mean. A **large SD** means values are widely spread.
In clinical practice, you may see lab reports written as: **Mean ± SD**, for example: Hemoglobin = **12.5 ± 1.8 g/dL**

Quartiles and the Interquartile Range (IQR)

The median splits the ordered data into two halves. **Quartiles** split it into four quarters:

- **Q1** (first quartile) — 25% of values lie below it
- **Q2** = the median — 50% of values lie below it
- **Q3** (third quartile) — 75% of values lie below it

The **interquartile range** is the spread of the middle 50% of the data:

$$\text{IQR} = Q_3 - Q_1$$

How to find Q1 and Q3 (the same rule SPSS uses in *Explore*):

1. Arrange the data in order.
2. Q1 is at position $0.25(n+1)$; Q3 is at position $0.75(n+1)$.
3. If the position is not a whole number, go that fraction of the way between the two neighbouring values.

Example: Length of Stay

Length of stay (days) of 11 patients, already in order:

$$2,\ 3,\ 3,\ 4,\ 4,\ 5,\ 5,\ 6,\ 7,\ 9,\ 21$$

- Q1: position $0.25(11+1) = 3$, so Q1 = **3 days**
- Median: position $0.50(11+1) = 6$, so median = **5 days**
- Q3: position $0.75(11+1) = 9$, so Q3 = **7 days**
- IQR $= 7 - 3 =$ **4 days**

Checking for outliers with the 1.5 × IQR rule:

- Upper fence = Q3 + 1.5 × IQR $= 7 + 6 = 13$ days
- Lower fence = Q1 − 1.5 × IQR $= 3 - 6 = -3$ days

The patient who stayed **21 days** is above 13, so this value is an **outlier**.

Example: When the Position Is Not a Whole Number

Number of clinic visits in one year for 12 patients, already in order:

$$3,\ 4,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9,\ 11,\ 12,\ 14,\ 20$$

- Q1: position $0.25(12+1) = 3.25$, so go 0.25 of the way from the 3rd value (4) to the 4th value (5): $Q_1 = 4 + 0.25(5-4) =$ **4.25**
- Median: position $0.50(12+1) = 6.5$, halfway between the 6th and 7th values: $\dfrac{7+8}{2} =$ **7.5**
- Q3: position $0.75(12+1) = 9.75$, so go 0.75 of the way from the 9th value (11) to the 10th value (12): $Q_3 = 11 + 0.75(12-11) =$ **11.75**
- IQR $= 11.75 - 4.25 =$ **7.5 visits**

Like the median, the IQR is **not affected by extreme values**, so the pair **median and IQR** is used to describe skewed data (the pair mean and SD is used for symmetric data).

Different Software, Slightly Different Quartiles

There are several accepted rules for finding quartiles, so a calculator, Excel, or another textbook may give a slightly different Q1 or Q3 for small samples. In this course we use the $0.25(n+1)$ and $0.75(n+1)$ positions, which match the SPSS *Explore* output.

Summary: Numerical Measures

Table 2: Summary of Numerical Summary Measures

Measure	Symbol	Purpose	Appropriate Data Type
Mean	xbar	Central tendency	Numerical
Median	Md	Central tendency	Numerical / Ordinal
Mode	Mo	Central tendency	Any
Range	Max - Min	Spread	Numerical
Variance	s2	Spread (squared units)	Numerical
Standard Deviation	s	Spread (original units)	Numerical
Interquartile Range	Q3 - Q1	Spread of the middle 50%	Numerical / Ordinal

Example 1: Calculating Mean, Median, and SD

A health worker records the resting heart rate (beats per minute) of **9 patients**:

$$72,\ 68,\ 85,\ 74,\ 90,\ 68,\ 78,\ 72,\ 110$$

Calculate the **(a) mean**, **(b) median**, and **(c) standard deviation**.

Solution

Step 1 — Arrange data in order:

$$68,\ 68,\ 72,\ 72,\ 74,\ 78,\ 85,\ 90,\ 110$$

(a) Mean:

$$\bar{x} = \frac{68 + 68 + 72 + 72 + 74 + 78 + 85 + 90 + 110}{9} = \frac{717}{9} = 79.7 \text{ bpm}$$

(b) Median:

$n = 9$ (odd), so median is at position $\dfrac{9+1}{2} = 5$th value.

Counting from the ordered list: the 5th value = **74 bpm**
Notice that the median (74) is lower than the mean (79.7) because the unusually high value of 110 pulls the mean upward.

(c) Standard Deviation:
Make a table with x and x^2, and add up each column:

Table 3: Column totals for the standard deviation

Patient	Heart rate x	x^2
Patient 1	68	4,624
Patient 2	68	4,624
Patient 3	72	5,184
Patient 4	72	5,184
Patient 5	74	5,476
Patient 6	78	6,084
Patient 7	85	7,225
Patient 8	90	8,100
Patient 9	110	12,100
Total	**717**	**58,601**

With $n = 9$: $\sum x = 717$ and $\sum x^2 = 58{,}601$.

$$n\sum x^2 - \left(\sum x\right)^2 = 9(58{,}601) - 717^2 = 527{,}409 - 514{,}089 = 13{,}320$$

$$s^2 = \frac{13{,}320}{9(9-1)} = \frac{13{,}320}{72} = 185$$

$$s = \sqrt{185} = 13.60 \text{ bpm}$$

Interpretation: The average heart rate is **79.7 ± 13.6 bpm**. The relatively large SD reflects the wide spread, largely influenced by the patient with 110 bpm.

Pictorial Summarization

Charts and graphs turn numbers into visual stories. The correct chart depends on the **data type**.

Bar Chart

Used for **categorical data** (nominal or ordinal). Each bar represents a category; the height shows the count or percentage.

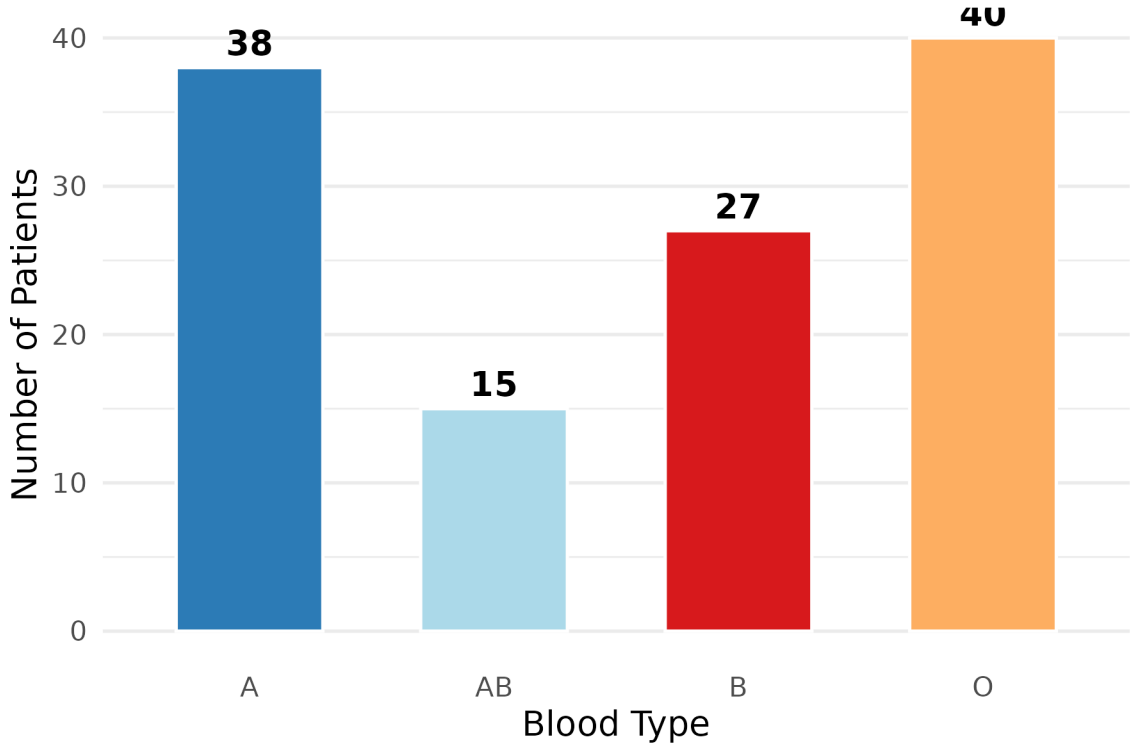


Figure 1: Bar Chart: Blood Types of 120 Patients in a Medical Ward

Histogram

Used for **continuous numerical data**. Bars are drawn over class intervals (ranges of values); bars touch each other because the data is continuous. *(We will learn how to choose the class intervals in Chapter 3.)*

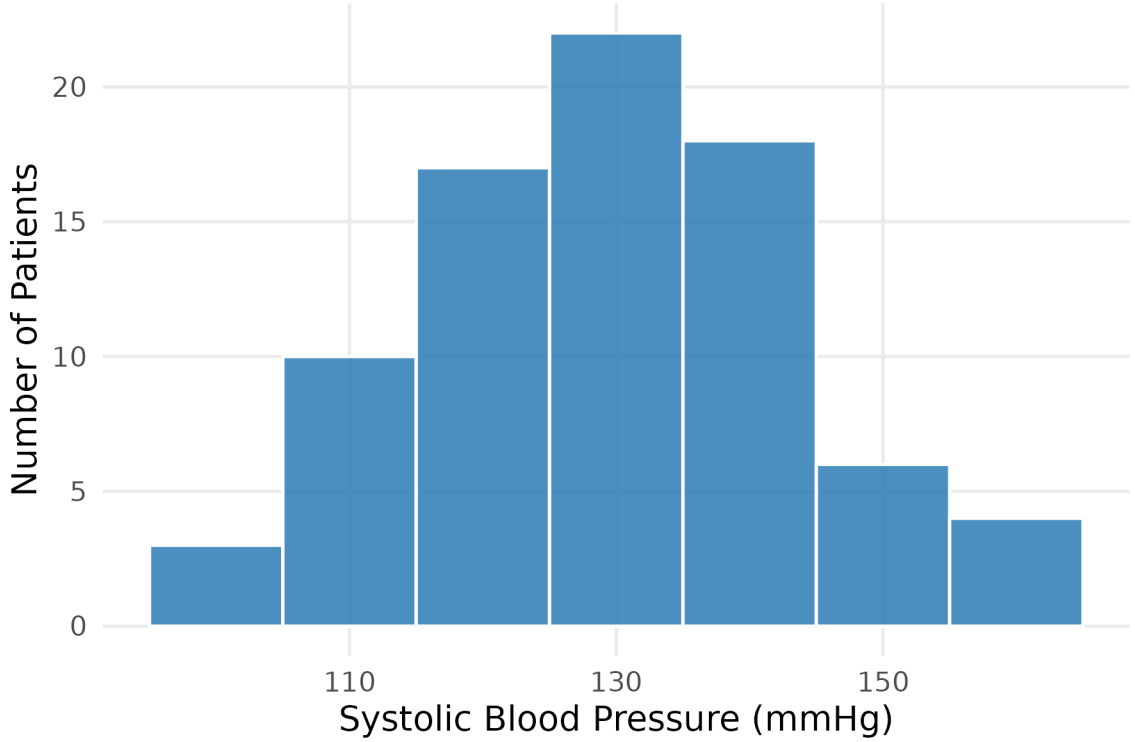


Figure 2: Histogram: Systolic Blood Pressure of 80 Patients

Bar Chart vs Histogram

	Bar Chart	Histogram
Data type	Categorical, or discrete with few values	Continuous numerical
Bars	Separated (gaps)	Touching (no gaps)
X-axis	Categories, or each possible count	Numerical ranges

Discrete data: if a count has only a **few possible values** (e.g., number of falls per month: 0, 1, 2, …, 5), draw a **bar chart** with one bar for each value. The gaps between the bars show that there are no values in between (no month has 1.5 falls). If a count has **many possible values** (e.g., number of patients seen per day, 0 to 80), group the values into class intervals and draw a **histogram** (Chapter 3).
A histogram can have gaps too. A class with frequency 0 leaves an empty space in a histogram. It is still a histogram, because its x-axis is a **number line** with every class shown in order, while a bar chart's x-axis is a list of separate categories.

Box Plot (Box-and-Whisker Plot)

A **box plot** summarizes the **distribution** of numerical data using 5 key values (the **five-number summary**):

Value	Description
Minimum	Smallest value (excluding outliers)
Q1	25th percentile (lower quartile)
Median (Q2)	50th percentile — middle value
Q3	75th percentile (upper quartile)
Maximum	Largest value (excluding outliers)

The **Interquartile Range (IQR) = Q3 − Q1** represents the middle 50% of data. Points beyond **1.5 × IQR** from Q1 or Q3 are marked as **outliers** (drawn as separate circles), as in the length-of-stay example above.

Shape of a Distribution: Symmetric or Skewed?

A distribution is **right-skewed** when it has a long tail of large values (like length of stay), and **left-skewed** when the long tail is of small values. Four ways to tell:

- **Box plot — look at the box**, not only the whiskers. If the median is closer to Q_1 (the part from the median to Q_3 is longer), the data are right-skewed; if it is closer to Q_3, left-skewed.
- **Histogram:** the long tail is on the right (right-skewed) or on the left (left-skewed).
- **Mean and median:** usually mean > median when right-skewed, and mean < median when left-skewed.
- **SPSS Skewness:** a value > 0 means right-skewed, < 0 means left-skewed, close to 0 means roughly symmetric.

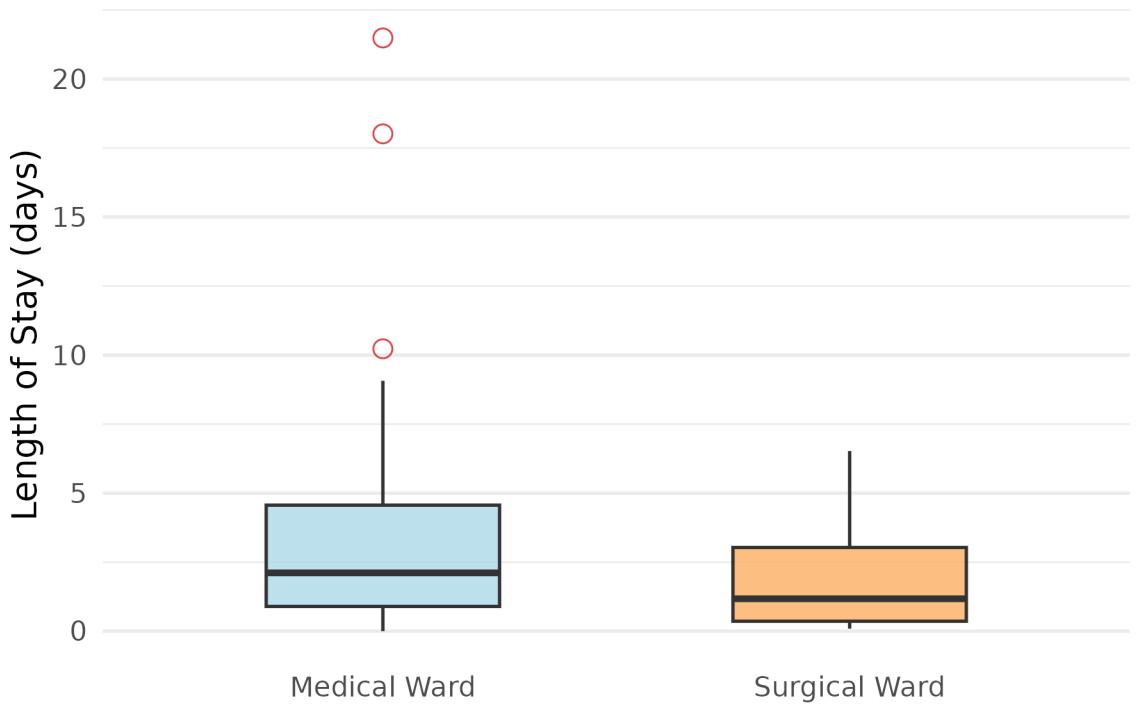


Figure 3: Box Plot: Comparing Length of Hospital Stay Between Two Wards

Example (clinic visits, above): median $- Q_1 = 7.5 - 4.25 = 3.25$ is smaller than $Q_3 -$ median $= 11.75 - 7.5 = 4.25$, the mean (8.58) is larger than the median (7.5), and SPSS gives Skewness = 1.12, so the data are **right-skewed**. Both wards in the box plot above are right-skewed too.

Choosing the Right Chart

Table 1: Guide to Choosing the Right Chart

Chart Type	Data Type	What It Shows	Example
Bar Chart	Categorical (Nominal/Ordinal); Discrete with few values	Frequency or percentage of each category	Number of patients by blood type; number of falls per month (0-5)
Pie Chart	Categorical (Nominal) - proportions	Proportion of each category as a slice of the whole	Proportion of patients by diagnosis

Histogram	Numerical (Continuous); Discrete with many values	Distribution shape, spread, and central tendency	Distribution of patient ages or blood pressure
Box Plot	Numerical (Continuous/Discrete)	Median, spread, quartiles, and outliers	Comparing recovery time between two wards

---

Example 2: Choosing the Right Summary and Chart

A hospital administrator records the **diagnosis** of 200 patients admitted last month and wants to summarize the data. A researcher records the **length of stay (days)** for the same 200 patients.

Solution

Variable	Data Type	Best Summary Measure	Best Chart
Diagnosis	Categorical – Nominal	Mode (most common diagnosis)	Bar Chart
Length of stay	Numerical – Continuous (likely skewed)	Median + IQR	Box Plot or Histogram

Why median for length of stay? Hospital stays are almost always right-skewed — most patients stay a few days, but a small number may stay for weeks or months. These long-stay patients would pull the mean upward, making it misleading. The median better represents the "typical" patient.

Key Takeaways

Chapter 2 Summary

1. **Numerical summarization** uses measures of *central tendency* (mean, median, mode) and *variability* (range, IQR, variance, SD) to describe data with numbers.
2. The **mean** is best for symmetric numerical data; the **median** is better when data is skewed or has outliers; the **median** is also used for ordinal data, and the **mode** for nominal data.
3. The **standard deviation (SD)** is the most commonly reported measure of spread in clinical research, expressed in the same units as the data.
4. **Pictorial summarization** uses charts: bar charts for categorical data (and discrete counts with few values), histograms for continuous data, and box plots to compare groups and identify outliers.
5. Always choose the **correct summary measure and chart** based on the **data type** — using the wrong one leads to misleading conclusions.

Formula Summary

Table 1: Formulas in this chapter

Quantity	Formula	Notes
Sample mean	$\bar{x} = \dfrac{\sum x_i}{n}$	Symmetric numerical data
Median	Value at position $\dfrac{n+1}{2}$ in the ordered data	n even: average of the two middle values
Range	Maximum − Minimum	Uses only two values
Population variance	$\sigma^2 = \dfrac{\sum (x_i - \mu)^2}{N}$	Whole population of size N

Quantity	Formula	Notes
Sample variance	$s^2 = \dfrac{n\sum x^2 - \left(\sum x\right)^2}{n(n-1)}$	Squared units; needs $\sum x$ and $\sum x^2$
Sample SD	$s = \sqrt{s^2}$	Same units as the data; report Mean $\pm$ SD
Quartiles	Q_1 at position $0.25(n+1)$, Q_3 at $0.75(n+1)$	Same rule as SPSS Explore
Interquartile range	$\text{IQR} = Q_3 - Q_1$	Report Median and IQR for skewed data
Outlier fences	$Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$	Values outside are outliers on a box plot

Exercises

Exercise 1

A health worker records the body temperature (°C) of **10 patients** at 8:00 AM:

$$37.2,\ 36.8,\ 38.5,\ 37.0,\ 39.1,\ 36.8,\ 37.5,\ 38.0,\ 37.2,\ 40.2$$

(a) Calculate the **mean** body temperature.
(b) Find the **median** body temperature.
(c) What is the **mode**?
(d) Calculate the **range**.
(e) Make a table with columns x and x^2, then use the computational formula to calculate the **standard deviation**. Show all steps.
(f) One patient has a temperature of 40.2°C. How does this value affect the mean compared to the median? The ward's infection-control team wants a single number for the *typical* temperature on the ward this morning. Which measure would you give them, and why? *(The patient with 40.2°C must, of course, still be reported to the physician individually.)*

Exercise 2

A community health worker surveys **50 elderly patients** and records:

- **Gender** (Male / Female)
- **Systolic blood pressure** (mmHg)
- **Self-rated health** (Poor / Fair / Good / Very Good / Excellent)

For each variable, state:
(a) The data type.
(b) The most appropriate measure of central tendency.
(c) The most appropriate chart to display the data.

Exercise 3

The number of patient falls recorded each month over **12 months** in a hospital ward are:

$$2,\ 0,\ 3,\ 1,\ 4,\ 2,\ 1,\ 0,\ 3,\ 2,\ 5,\ 1$$

(a) Calculate the mean and median number of falls per month.
(b) Count how many months had 0, 1, 2, 3, 4 and 5 falls. Draw a **bar chart** (by hand or describe what it would look like) with the number of falls on the x-axis and the number of months on the y-axis. Why is a bar chart, not a histogram, suitable here?
(c) The ward manager wants to set a monthly target. Should she use the mean or median as the baseline? Explain.

End of Chapter 2

Coming Up Next

Chapter 3: Random Variable and Probability Distribution — We will learn what a random variable is, how probabilities work, and why the **normal distribution** (bell curve) is so important in health science research.

Random Variable and Probability Distribution

Introduction

Why Does This Matter?

When we collect data — patient ages, blood pressure readings, test results — we rarely get the same value twice. Outcomes vary, and that variation follows patterns we can describe mathematically.
Understanding **how data is distributed** allows us to answer questions like:

- What is the probability that a randomly selected patient has a fever above 39°C?
- What proportion of adults have a BMI in the healthy range?
- How unusual is a blood pressure reading of 160 mmHg in this population?

In this chapter, we start from **frequency distributions** (what we observed in a sample), move to **probability** and **random variables** (what we expect in the population), and finish with the **normal distribution** — the most important probability distribution in all of biostatistics.

Frequency Distributions

What Is a Frequency Distribution?

A **frequency distribution** is a systematic way of organizing raw data into a table that shows how often each value (or range of values) occurs.

It answers the simple question: *"How many times does each value appear?"*

Frequency Distribution for Categorical Data

For categorical variables, we simply count how many observations fall into each category.

Example

Blood types of 200 patients admitted to a hospital last month:

Table 1: Frequency Distribution: Blood Type of 200 Patients

Blood Type	Frequency (f)	Relative Frequency (f/n)	Percentage
A	76	$76/200 = 0.38$	38%
B	54	$54/200 = 0.27$	27%
AB	30	$30/200 = 0.15$	15%
O	40	$40/200 = 0.20$	20%
Total	**200**	**$200/200 = 1.00$**	**100%**

Relative frequency $= \dfrac{f}{n}$, where f is the count for that category and n is the total number of observations.

Frequency Distribution for Continuous Data

For continuous variables, we group values into **class intervals** (also called *bins*) and count how many observations fall in each interval.

Steps to construct a frequency distribution table:

1. Find the **range** = Maximum − Minimum
2. Choose the **number of classes** (typically 5–10)
3. Calculate **class width** = $\dfrac{\text{Range}}{\text{Number of classes}}$ (round up)
4. List the class intervals and count observations in each

Example

Systolic blood pressure (mmHg) of 80 patients:

Table 2: Frequency Distribution: Systolic Blood Pressure of 80 Patients

Class Interval (mmHg)	Frequency (f)	Relative Frequency	Cumulative Frequency
95-104	2	0.025	2
105-114	9	0.112	11
115-124	17	0.212	28
125-134	22	0.275	50
135-144	18	0.225	68
145-154	8	0.100	76
155-164	4	0.050	80
165-174	0	0.000	80

Cumulative frequency tells us how many observations fall *at or below* the upper boundary of each class — useful for answering questions like *"What proportion of patients have SBP below 135 mmHg?"* (read the cumulative frequency of the class that ends at 134, then divide by 80).

The cumulative column also gives the **median**: it is the first value (or class) whose **cumulative percent reaches 50%**. Here 50% of 80 is 40 patients, and the first class whose cumulative frequency reaches 40 is **125-134 mmHg**, so that class contains the median. In SPSS *Frequencies* output, read down the **Cumulative Percent** column in the same way.

Example 1: Constructing a Frequency Distribution Table

The following are the ages (in years) of 20 patients:

$$45, 62, 38, 55, 71, 48, 60, 33, 52, 67,$$
$$41, 58, 74, 50, 65, 44, 57, 69, 36, 53$$

Construct a frequency distribution table using **5 class intervals**.

Solution

Step 1 — Find the range:

$$\text{Range} = 74 - 33 = 41$$

Step 2 — Calculate class width:

$$\text{Class width} = \frac{41}{5} = 8.2 \rightarrow \text{round up to } 9$$

Step 3 — Construct the table (starting from the minimum, 33). Each class is 9 years wide, so the classes are 33–41, 42–50, 51–59, 60–68 and 69–77. Every value belongs to **exactly one** class — for example, age 60 is counted in 60–68, not in 51–59.

Table 3: Frequency Distribution of Patient Ages (n = 20)

Age Group (years)	Frequency (f)	Relative Frequency	Percentage	Cumulative Frequency
33-41	4	0.20	20%	4
42-50	4	0.20	20%	8
51-59	5	0.25	25%	13
60-68	4	0.20	20%	17
69-77	3	0.15	15%	20

Interpretation: The largest group of patients is aged **51–59 years** (5 patients, 25%). From the cumulative frequency column, 8 of the 20 patients (40%) are aged 50 years or younger, and 13 patients (65%) are aged 59 years or younger.

Visualizing Frequency Distributions

A frequency distribution table can be visualized as a **histogram**, where the height of each bar represents the frequency of that class interval.

The **shape** of the histogram is important — it tells us about the **distribution** of the data, which leads us to the next section.

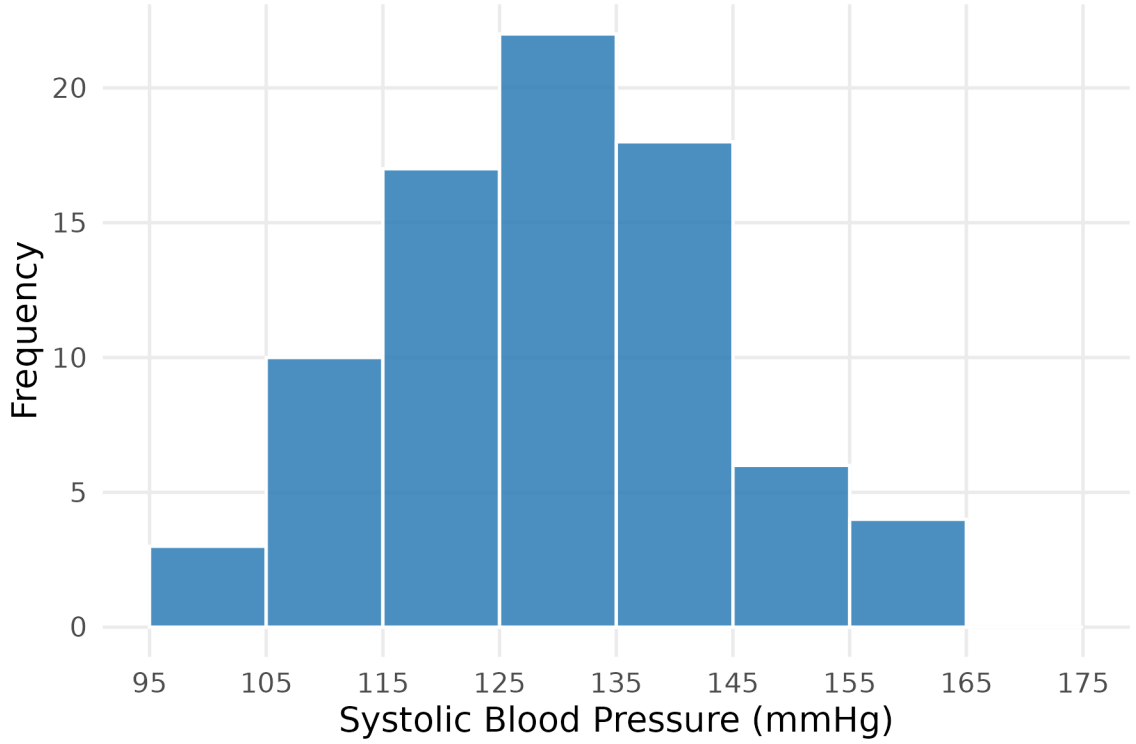


Figure 1: Histogram of Systolic Blood Pressure (80 patients)

From Frequency to Probability

Probability

A **probability** is a number between 0 and 1 that describes how likely an event is:

- $P = 0$ — the event never happens
- $P = 1$ — the event always happens
- $P = 0.25$ — the event happens about 25% of the time in the long run

The **relative frequency** we calculated above is our link to probability. In the blood-type table, 38% of the 200 patients had blood type A. If we pick one patient **at random** from this group, the probability that the patient has blood type A is $P(\text{type A}) = 0.38$.

When we only have a sample, the relative frequency is our best *estimate* of the probability in the population.

Two rules for questions about more than one category:

- **"A or B"** (categories that cannot happen together): **add** their probabilities. $P(\text{type A or type AB}) = 0.38 + 0.15 = 0.53$, because no patient has two blood types.
- **"At least"**: use the cumulative column and the complement, $P(\text{at least } k) = 1 - P(\text{at most the value below } k)$. From the SBP table, $P(\text{SBP} \geq 135) = 1 - P(\text{SBP} \leq 134) = 1 - \frac{50}{80} = 0.375$.

Random Variable

A **random variable** is a variable whose value depends on **chance** — we do not know its value until we select an individual and measure it.

> **Example:** "The systolic blood pressure of a randomly selected patient" is a random variable. Before we choose the patient, we cannot say what the value will be, but we *can* say which values are more likely.

Random variables follow the same classification as data in Chapter 1:

Random variable	Example	Probabilities are shown as
Discrete	Number of falls on a ward per month	A bar for each possible count
Continuous	Systolic blood pressure	The **area under a smooth curve**

A **probability distribution** lists all possible values of a random variable and how likely each one is.

For a continuous random variable, imagine collecting more and more patients and making the class intervals of the histogram narrower and narrower: the bars smooth out into a **curve**. The probability that a value falls in a range equals the **area under the curve** over that range, and the **total area is 1** (100%).

The most important of these curves is the **normal distribution**.

The Normal Distribution

What Is the Normal Distribution?

The **normal distribution** (also called the *Gaussian distribution* or *bell curve*) is the most important probability distribution in statistics.

Many biological and clinical measurements naturally follow — or approximately follow — a normal distribution:

- Adult height
- Blood pressure in a healthy population
- Hemoglobin level in healthy adults
- Birth weight of full-term newborns

Not every clinical measurement is normal. Length of hospital stay, for example, is strongly **right-skewed** (Chapter 2), so it does not follow a normal distribution.

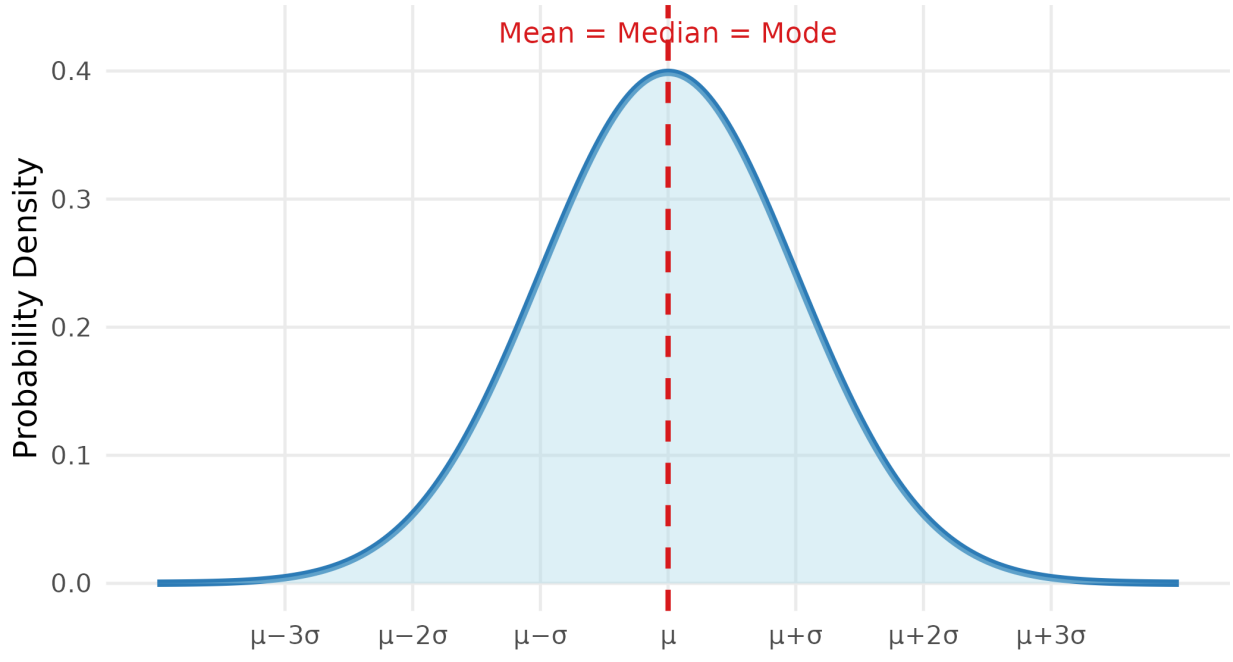


Figure 1: The Normal Distribution: a symmetric, bell-shaped curve

Key Properties of the Normal Distribution

A normal distribution is completely described by just **two parameters**:

Parameter	Symbol	Meaning
Mean	μ	Center of the distribution (peak of the bell)
Standard deviation	σ	Width of the bell — larger = wider, flatter curve

Key properties:

1. **Symmetric** around the mean — the left and right halves are mirror images
2. **Mean = Median = Mode** — all three are equal and located at the center
3. The **total area** under the curve = 1 (representing 100% of all probability)
4. The curve extends infinitely in both directions but never touches the x-axis

The 68–95–99.7 Rule (Empirical Rule)

One of the most useful facts about the normal distribution is how much of the data falls within 1, 2, or 3 standard deviations of the mean.

The Empirical Rule

Range	Proportion of data
Within **1 SD** of the mean: $\mu \pm \sigma$	**68%**
Within **2 SD** of the mean: $\mu \pm 2\sigma$	**95%**
Within **3 SD** of the mean: $\mu \pm 3\sigma$	**99.7%**

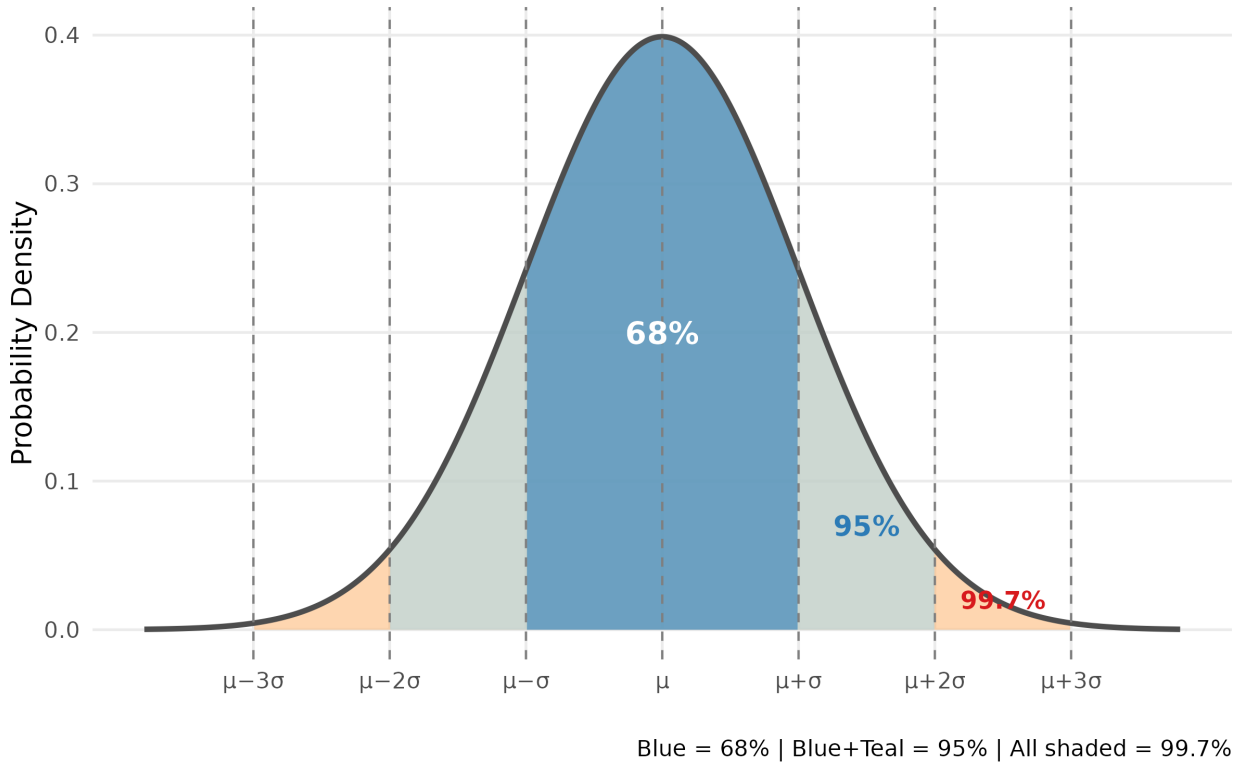


Figure 2: The 68–95–99.7 Rule for the Normal Distribution

This means only about **5%** of values fall more than 2 SD from the mean, and only **0.3%** fall more than 3 SD away — these are considered unusual or extreme.
More precisely, 95% of values lie within **1.96** *SD of the mean. The rounded value 2 is fine for quick estimates; the exact value 1.96 is the one we will use for confidence intervals in Chapter 4.*

Example 2: Applying the Empirical Rule

The systolic blood pressure (SBP) of adults in a community follows a normal distribution with **mean = 125 mmHg** and **SD = 14 mmHg**.

(a) Between what two values does the SBP of the **middle 95%** of adults fall?

(b) What percentage of adults have SBP **above 153 mmHg**?

Solution

(a) The middle 95% falls within $\mu \pm 2\sigma$:

$$125 - 2(14) = 97 \text{ mmHg} \quad \text{to} \quad 125 + 2(14) = 153 \text{ mmHg}$$

So **95% of adults** have SBP between **97 and 153 mmHg**.
(b) SBP = 153 mmHg is exactly $\mu + 2\sigma$. Since 95% of values fall *within* 2 SD, the remaining 5% fall *outside*. By symmetry, **2.5%** fall above 153 mmHg and 2.5% fall below 97 mmHg.
Answer: Approximately **2.5%** of adults have SBP above 153 mmHg.

The Standard Normal Distribution and Z-Score

The Z-Score

Different normal distributions have different means and SDs. To compare values across different distributions, we **standardize** them using the **Z-score**:

$$z = \frac{x - \mu}{\sigma}$$

Where:

- x = the observed value
- μ = the population mean
- σ = the population standard deviation
- z = the number of standard deviations x is away from the mean

A **positive Z** means the value is above the mean; a **negative Z** means it is below the mean.

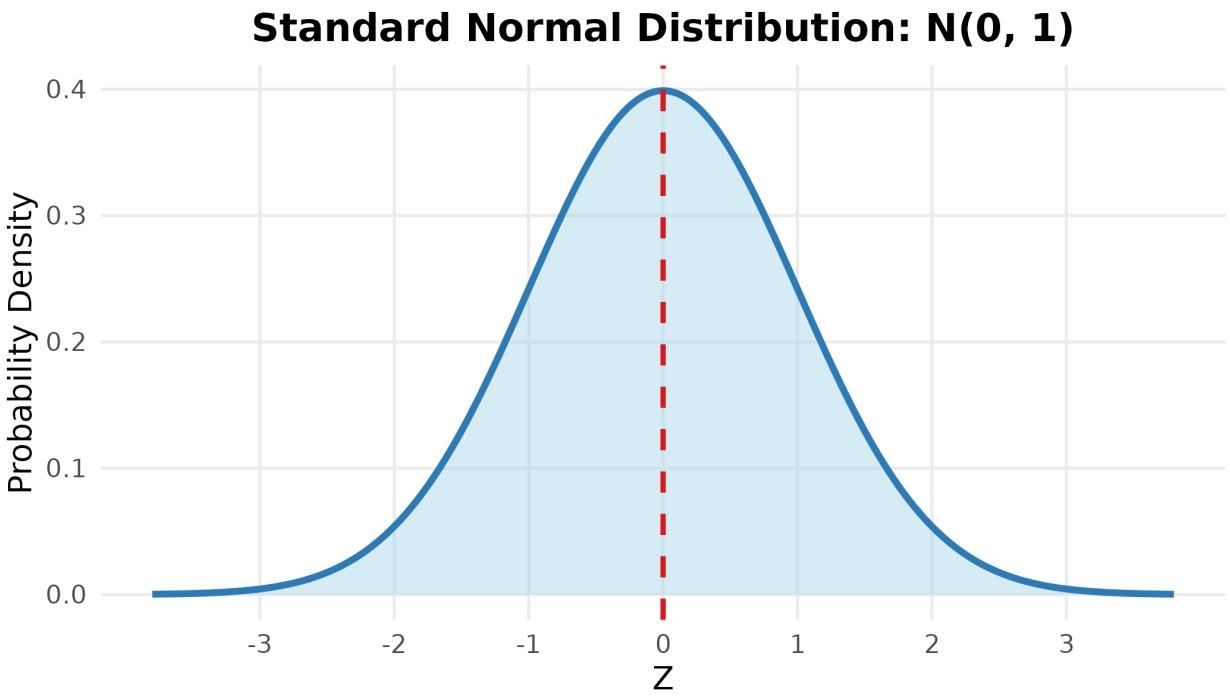


Figure 3: Standard Normal Distribution N(0,1)

The Standard Normal Distribution

When we convert all values to Z-scores, the resulting distribution is called the **standard normal distribution**, with:

$$\mu = 0 \quad \text{and} \quad \sigma = 1$$

This is written as $Z \sim N(0, 1)$.

Finding Probabilities Using the Z-Table

The **Z-table** (also called the standard normal table) gives us the **area to the left** of any Z-score — which equals the **probability** that a randomly selected observation falls below that value.

Table 1: Area to the left of Z (rows = Z integer + 1st decimal, columns = 2nd decimal)

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359

0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

How to use the table: To find $P(Z < 1.23)$, look at row **1.2** and column **0.03**, then read off the value: $P(Z < 1.23) = 0.8907$.

Negative Z-Scores: Use Symmetry

The table above only lists **positive** Z-scores. Because the normal curve is **symmetric** around 0, the area to the left of $-z$ is the same as the area to the right of $+z$:

$$P(Z < -z) = P(Z > z) = 1 - P(Z < z)$$

Example: $P(Z < -1.50) = 1 - P(Z < 1.50) = 1 - 0.9332 = 0.0668$

Common Z-Table Probability Rules

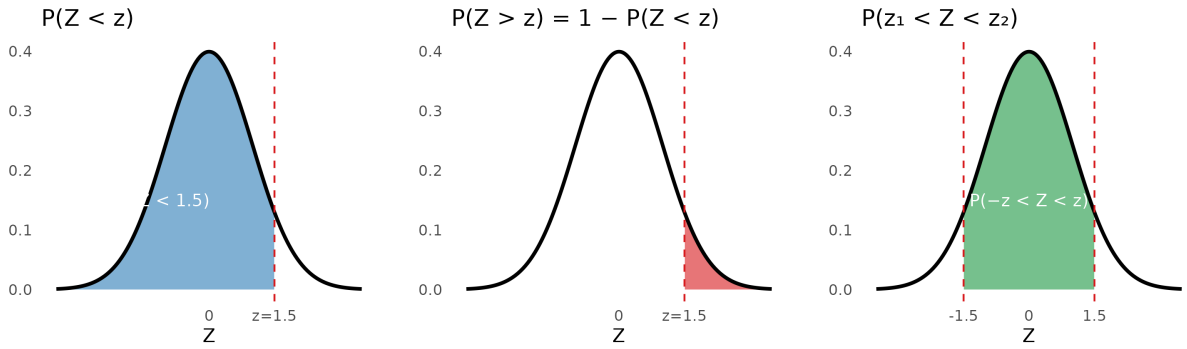


Figure 4: Three useful probability rules for the normal distribution

Key Probability Rules

Situation	Formula
Probability **below** z	$P(Z < z)$ — read directly from Z-table
Probability **above** z	$P(Z > z) = 1 - P(Z < z)$
Probability **between** two values	$P(z_1 < Z < z_2) = P(Z < z_2) - P(Z < z_1)$
Probability below a **negative** z	$P(Z < -z) = 1 - P(Z < z)$
Value with a given area below it	find the area **inside** the table, read z, then $x = \mu + z\sigma$

Example 3: Using Z-Scores and the Z-Table

Body weight of adult males in a region is approximately normally distributed with **mean = 70 kg** and **SD = 10 kg**.

(a) What is the probability that a randomly selected adult male weighs **less than 85 kg**?

(b) What is the probability that he weighs **more than 55 kg**?

(c) What is the probability that he weighs **between 60 and 80 kg**?

Solution

(a) P(X < 85):

$$z = \frac{85 - 70}{10} = \frac{15}{10} = 1.50$$

From Z-table: $P(Z < 1.50) = \mathbf{0.9332}$
93.3% of adult males weigh less than 85 kg.

(b) P(X > 55):

$$z = \frac{55 - 70}{10} = \frac{-15}{10} = -1.50$$

Using the symmetry rule, $P(Z < -1.50) = 1 - 0.9332 = 0.0668$, so

$$P(Z > -1.50) = 1 - P(Z < -1.50) = 1 - 0.0668 = \mathbf{0.9332}$$

93.3% of adult males weigh more than 55 kg. *(This makes sense by symmetry — 55 and 85 are equidistant from the mean.)*

(c) P(60 < X < 80):

$$z_1 = \frac{60 - 70}{10} = -1.00 \qquad z_2 = \frac{80 - 70}{10} = 1.00$$

Using symmetry, $P(Z < -1.00) = 1 - 0.8413 = 0.1587$, so

$$P(-1.00 < Z < 1.00) = P(Z < 1.00) - P(Z < -1.00)$$
$$= 0.8413 - 0.1587 = \mathbf{0.6826}$$

68.3% of adult males weigh between 60 and 80 kg — consistent with the 68–95–99.7 rule.

Reading the Z-Table Backwards: From an Area to z

So far we went from a value to a probability: $x \to z \to$ area. Sometimes we know the **area** and need the **value** — for example, the cut-off below which the lowest 5% of patients fall.

1. Look for the area (area to the left) **inside the body** of the Z-table, then read z from its row and column. If it lies between two entries, use the closer one.

- 0.95 lies between 0.9495 ($z = 1.64$) and 0.9505 ($z = 1.65$), so $z_{0.95} = 1.645$.
- 0.90 is closest to 0.8997, so $z_{0.90} = 1.28$.

2. For an area **below 0.5**, use symmetry: $z_{0.05} = -z_{0.95} = -1.645$.
3. Change z back to the original units: $x = \mu + z\sigma$.

Example 4: From a Percentage Back to a Value

The hemoglobin level of healthy adult women is approximately normally distributed with **mean = 13.5 g/dL** and **SD = 1.0 g/dL**.

(a) Below what hemoglobin level do the lowest **5%** of women fall?

(b) What hemoglobin level is the **90th percentile**?

Solution

(a) Area to the left = 0.05, so $z_{0.05} = -1.645$.

$$x = \mu + z\sigma = 13.5 + (-1.645)(1.0) = \mathbf{11.86} \text{ g/dL}$$

The lowest 5% of healthy women have hemoglobin below about 11.9 g/dL (close to the 12 g/dL commonly used to define anemia in women).

(b) Area to the left = 0.90, so $z_{0.90} = 1.28$.

$$x = 13.5 + 1.28(1.0) = \mathbf{14.78} \text{ g/dL}$$

The Same Calculations in SPSS

In **Transform → Compute Variable**:

- `CDF.NORMAL(x, mean, sd)` $= P(X < x)$, the area to the left of x. For an area to the right, use 1− `CDF.NORMAL(x, mean, sd)`.
- `IDF.NORMAL(p, mean, sd)` does the reverse: the value with area p to its left, e.g. `IDF.NORMAL(0.05, 13.5, 1)` = 11.86.

SPSS rounds probabilities to 3 decimals. A value shown as **.000** means less than .0005, **not zero** — report it as $p < .001$.

Key Takeaways

Chapter 3 Summary

1. A **frequency distribution** organizes data into a table showing how often each value or range of values occurs. It can show absolute frequency, relative frequency, or cumulative frequency.

2. The **normal distribution** is a symmetric, bell-shaped distribution described by its mean (μ) and standard deviation (σ).

3. A **probability** is a number from 0 to 1; for a continuous **random variable** it is the **area under the curve**. The **68–95–99.7 rule**: approximately 68%, 95%, and 99.7% of values fall within 1, 2, and 3 standard deviations of the mean, respectively.

4. A **Z-score** measures how many standard deviations a value is from the mean: $z = \dfrac{x - \mu}{\sigma}$

5. The **Z-table** gives the probability (area) to the left of any Z-score, allowing us to find the probability that a measurement falls in any range.

Formula Summary

Table 1: Formulas in this chapter

Quantity	Formula	Notes
Relative frequency	$\dfrac{f}{n}$	Multiply by 100 for a percentage
Class width	$\dfrac{\text{Range}}{\text{Number of classes}}$, rounded up	Usually 5 to 10 classes
Empirical rule	$\mu \pm \sigma$: 68%, $\mu \pm 2\sigma$: 95%, $\mu \pm 3\sigma$: 99.7%	Normal distribution only

Quantity	Formula	Notes
Z-score	$z = \dfrac{x - \mu}{\sigma}$	Number of SDs from the mean
Probability below	$P(Z < z)$	Read directly from the Z-table
Probability above	$P(Z > z) = 1 - P(Z < z)$	
Probability between	$P(z_1 < Z < z_2) = P(Z < z_2) - P(Z < z_1)$	
Negative Z	$P(Z < -z) = 1 - P(Z < z)$	Symmetry of the normal curve
Value from an area	$x = \mu + z\sigma$	Find the area inside the Z-table first; e.g. $z_{0.95} = 1.645$
A or B	$P(A \text{ or } B) = P(A) + P(B)$	Categories that cannot happen together
At least k	$1 - P(\text{at most the value below } k)$	Use the cumulative column

Exercises

Exercise 1

The fasting blood glucose levels (mg/dL) of 25 patients are:

$$95, 112, 88, 134, 102, 119, 145, 97, 108, 125,$$
$$91, 130, 103, 116, 88, 142, 99, 121, 107, 138,$$
$$94, 115, 126, 100, 111$$

(a) Construct a frequency distribution table with **5 class intervals**. Include columns for frequency, relative frequency, percentage, and cumulative frequency.
(b) Which class interval contains the most patients?
(c) Using the cumulative frequency column, what percentage of patients have a blood glucose level below 112 mg/dL?

Exercise 2

The birth weight of full-term newborns is approximately normally distributed with **mean = 3,300 g** and **SD = 400 g**.
Using the **68–95–99.7 rule**:
(a) What range of birth weights includes the **middle 68%** of newborns?
(b) What range includes the **middle 99.7%** of newborns?
(c) A newborn weighing less than **2,500 g** is classified as low birth weight. Based on the empirical rule, is this an unusual outcome? Explain.

Exercise 3

The resting heart rate of healthy adults follows a normal distribution with **mean = 72 bpm** and **SD = 8 bpm**.
Find the following probabilities using the Z-table. Show the Z-score calculation and draw a sketch of the normal curve for each part.
(a) $P(\text{heart rate} < 80)$
(b) $P(\text{heart rate} > 88)$
(c) $P(64 < \text{heart rate} < 88)$
(d) A heart rate below 60 bpm is considered **bradycardia**. What proportion of healthy adults would be classified as having bradycardia based on this distribution?

End of Chapter 3

Coming Up Next

Chapter 4: Estimation — Using what we know about the normal distribution, we will learn how to estimate population parameters from sample data and construct **confidence intervals** that quantify our uncertainty.

Estimation

Introduction

Why Does This Matter?

In practice, we almost never have access to an entire population. We collect data from a **sample** and use it to draw conclusions about the **population** — a process called **statistical inference**.
Estimation is one of the two main branches of statistical inference (the other being hypothesis testing, covered in Chapter 5).
In this chapter, we answer questions like:

- The mean serum cholesterol in our sample is 198 mg/dL — what can we say about the mean for the *entire* population?
- A survey found that 34% of respondents have hypertension — what is a reasonable range for the true population proportion?

The key tool for answering these questions is the **confidence interval**.

Concept of Estimation

Point Estimate vs. Interval Estimate

There are two ways to estimate a population parameter from sample data:

Point Estimate

A **point estimate** is a **single value** calculated from the sample that serves as our best guess for the population parameter.

Population Parameter	Symbol	Point Estimate	Symbol
Population mean	μ	Sample mean	$\bar{x}$
Population proportion	p	Sample proportion	$\hat{p}$
Population SD	σ	Sample SD	s

A point estimate is simple and easy to communicate, but it gives us **no information about how accurate or reliable** that single number is.

Interval Estimate (Confidence Interval)

An **interval estimate** gives a **range of plausible values** for the population parameter, along with a stated level of confidence.

$$\text{Point Estimate} \pm \text{Margin of Error}$$

This range is called a **confidence interval (CI)**.

> Key Concept: Confidence Level
>
> The **confidence level** (commonly **90%, 95%, or 99%**) tells us:
>
> *"If we repeated this study many times and calculated a confidence interval*

each time, approximately X% of those intervals would contain the true population parameter."

In health research, **95% confidence** is the most widely used standard.

The Sampling Distribution of the Mean

If we take **one** sample of 40 patients and calculate $\bar{x}$, we get one number. If a colleague takes a **different** sample of 40 patients from the same population, she will get a slightly different $\bar{x}$. The sample mean itself varies from sample to sample.

Imagine repeating this many times and drawing a histogram of all the sample means. This histogram is called the **sampling distribution of the mean**. It has three important features:

1. It is **centred at the population mean** μ.
2. Its spread is **smaller** than the spread of individual values: its standard deviation is $\sigma/\sqrt{n}$.
3. **Central Limit Theorem:** if the sample size is large enough (a common rule of thumb is $n \geq 30$), the sampling distribution of $\bar{x}$ is **approximately normal** — *even when the individual values are skewed.* If the individual values are already normal, it is normal for any n.

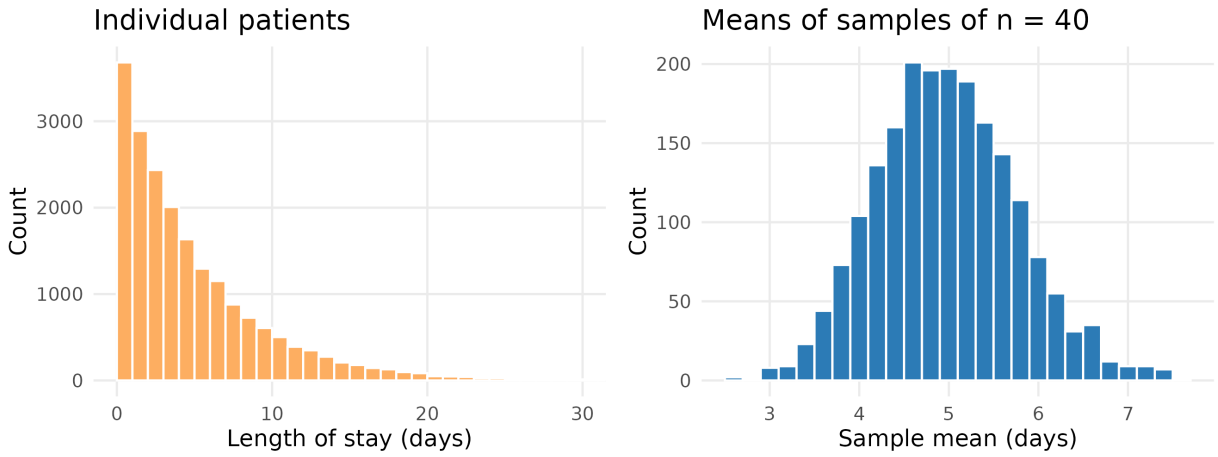


Figure 1: Left: length of stay of individual patients is right-skewed. Right: the means of many samples of 40 patients are approximately normal.

This is why we can use the **normal distribution** (Chapter 3) to build confidence intervals for a mean, and why the rule "$n \geq 30$" appears in the conditions below.

The Role of the Standard Error

The standard deviation of the sampling distribution has its own name: the **standard error (SE)**. It measures how much the sample estimate varies from sample to sample. Because we rarely know σ, we replace it with the sample SD s.

Standard Error of the Mean

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Where s is the sample standard deviation and n is the sample size.

Standard Error of a Proportion

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

SE vs. SD — What Is the Difference?

	Standard Deviation (SD)	Standard Error (SE)
Describes	Variability of individual observations	Variability of the sample mean
Gets smaller with	More homogeneous data	Larger sample size
Used for	Describing the spread of data	Constructing confidence intervals

The SE always gets smaller as n increases — larger samples give more precise estimates.

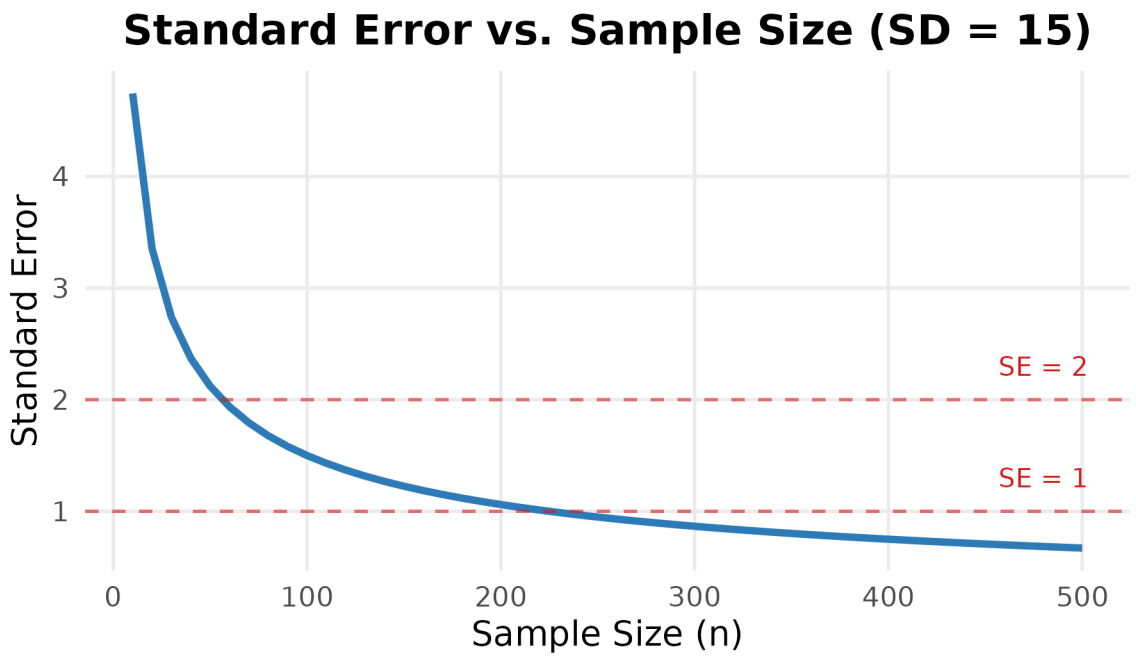


Figure 2: As sample size increases, the standard error decreases and estimates become more precise

Confidence Interval for a Mean

When the Population SD Is Known (Z-interval)

If the population standard deviation σ is known and either (a) the population is normal, or (b) the sample size is large ($n \geq 30$), we use the **Z-interval**:

$$\bar{x} \pm z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

The **critical value** $z_{1-\alpha/2}$ is written using the **left cumulative probability** (the area to the *left* of the value, matching the standard normal table and SPSS `IDF.NORMAL`). It depends on the confidence level:

Table 1: Critical Values for Common Confidence Levels

Confidence Level	alpha	1 - alpha/2 (left area)	z(1-alpha/2)
90%	0.10	0.950	1.645
95%	0.05	0.975	1.960
99%	0.01	0.995	2.576

The Most Used Value

In health research, the 95% CI with $z_{0.975} = 1.96$ is by far the most commonly reported. You will see it in almost every clinical paper.

When the Population SD Is Unknown (t-interval)

In practice, σ is almost never known. When we use the sample SD s as an estimate, we must use the **t-distribution** instead of the Z-distribution:

$$\bar{x} \pm t_{1-\alpha/2,\, n-1} \cdot \frac{s}{\sqrt{n}}$$

Where $t_{1-\alpha/2,\,n-1}$ is the critical value from the **t-distribution** with $n-1$ **degrees of freedom (df)**. As with z, the subscript $1-\alpha/2$ is the **left cumulative probability** (e.g., 0.975 for a 95% CI).

The t-Distribution

The t-distribution is similar to the standard normal distribution but has **heavier tails** — reflecting the extra uncertainty from estimating σ with s. As n increases, the t-distribution approaches the Z-distribution.

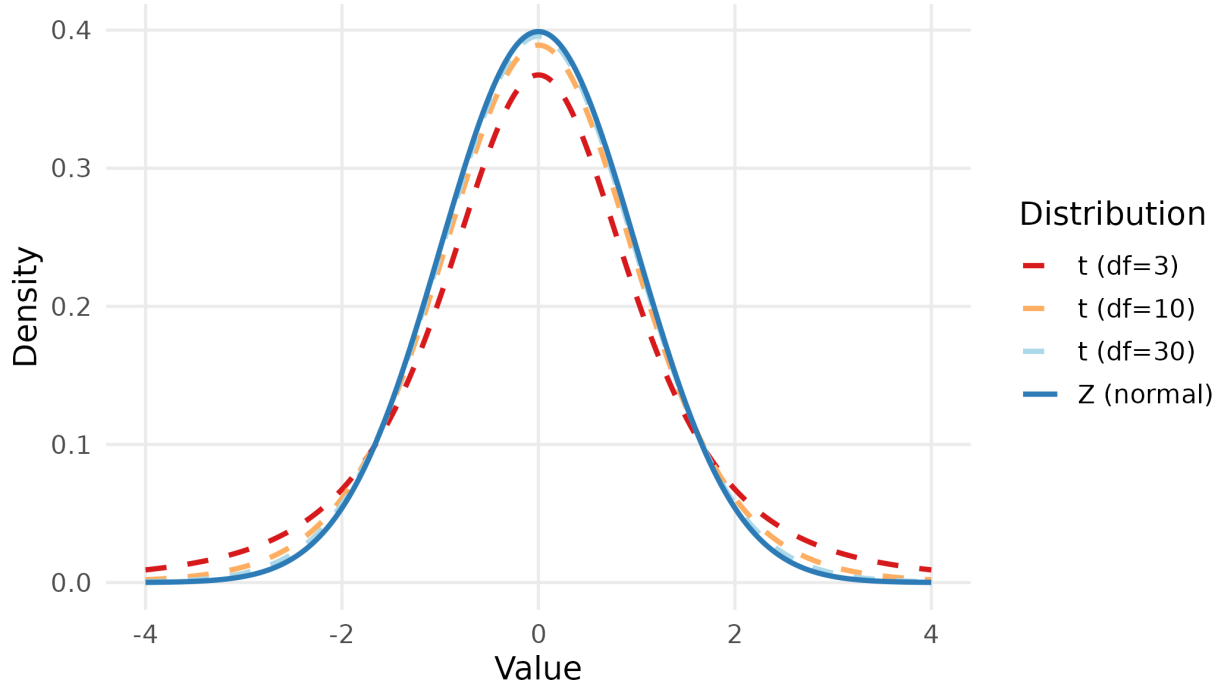


Figure 1: t-distributions with different degrees of freedom compared to Z

t Critical Values

Table 2: t Critical Values by left cumulative probability (use the 0.975 column for a 95% CI)

	Left cumulative probability			
df	0.95	0.975	0.99	0.995
1	6.314	**12.706**	31.821	63.657
2	2.920	**4.303**	6.965	9.925
3	2.353	**3.182**	4.541	5.841
4	2.132	**2.776**	3.747	4.604
5	2.015	**2.571**	3.365	4.032

6	1.943	**2.447**	3.143	3.707
7	1.895	**2.365**	2.998	3.499
8	1.860	**2.306**	2.896	3.355
9	1.833	**2.262**	2.821	3.250
10	1.812	**2.228**	2.764	3.169
11	1.796	**2.201**	2.718	3.106
12	1.782	**2.179**	2.681	3.055
13	1.771	**2.160**	2.650	3.012
14	1.761	**2.145**	2.624	2.977
15	1.753	**2.131**	2.602	2.947
16	1.746	**2.120**	2.583	2.921
17	1.740	**2.110**	2.567	2.898
18	1.734	**2.101**	2.552	2.878
19	1.729	**2.093**	2.539	2.861
20	1.725	**2.086**	2.528	2.845
21	1.721	**2.080**	2.518	2.831
22	1.717	**2.074**	2.508	2.819
23	1.714	**2.069**	2.500	2.807
24	1.711	**2.064**	2.492	2.797
25	1.708	**2.060**	2.485	2.787
26	1.706	**2.056**	2.479	2.779
27	1.703	**2.052**	2.473	2.771
28	1.701	**2.048**	2.467	2.763
29	1.699	**2.045**	2.462	2.756
30	1.697	**2.042**	2.457	2.750
35	1.690	**2.030**	2.438	2.724
40	1.684	**2.021**	2.423	2.704
50	1.676	**2.009**	2.403	2.678
60	1.671	**2.000**	2.390	2.660
70	1.667	**1.994**	2.381	2.648
80	1.664	**1.990**	2.374	2.639
100	1.660	**1.984**	2.364	2.626
120	1.658	**1.980**	2.358	2.617
Inf (Z)	1.645	**1.960**	2.326	2.576

How to use the table: for a 95% CI, use the column **0.975** and the row $df = n - 1$. For example, with $n = 25$: $t_{0.975,\,24} = 2.064$.

If your df is **not listed**, use the **next smaller** df in the table. This gives a slightly larger critical value, so the interval is slightly wider — the safe direction.

As df increases, the t critical value converges to **1.960** (the Z value). For $n \geq 30$ the difference is small, which is why some textbooks use Z as an approximation. **SPSS always uses t** when σ is unknown, and so do we in this course.

Interpreting a Confidence Interval

Correct Interpretation

> *"We are 95% confident that the true population mean lies between [lower bound] and [upper bound]."*

What this means: If we repeated the sampling process many times, 95% of the resulting confidence intervals would contain the true parameter.
What this does NOT mean: There is NOT a 95% probability that the true parameter lies in this specific interval — the true parameter is fixed; it either is or is not in the interval.

Using a CI to Judge a Target Value

A 95% CI contains the **plausible** values of the parameter. If a target or claimed value (for example, a hospital's target readmission rate) lies **outside** the interval, the data suggest the true value differs from that target. We will formalise this idea as a hypothesis test in Chapter 5.

The figure above shows 50 confidence intervals from 50 different samples of the same population. Most (95%) contain the true mean; a few do not — this is expected and not an error.

Example 1: 95% CI for a Mean (unknown)

A study measured the fasting blood glucose (mg/dL) of a random sample of **36 adults** and found:

$$\bar{x} = 105.4 \text{ mg/dL}, \qquad s = 18.6 \text{ mg/dL}$$

Construct a **95% confidence interval** for the population mean blood glucose.

Solution

Given: $n = 36$, $\bar{x} = 105.4$, $s = 18.6$, confidence level = 95%
Step 1 — Standard Error:

$$SE = \frac{s}{\sqrt{n}} = \frac{18.6}{\sqrt{36}} = \frac{18.6}{6} = 3.10$$

Step 2 — Critical value:

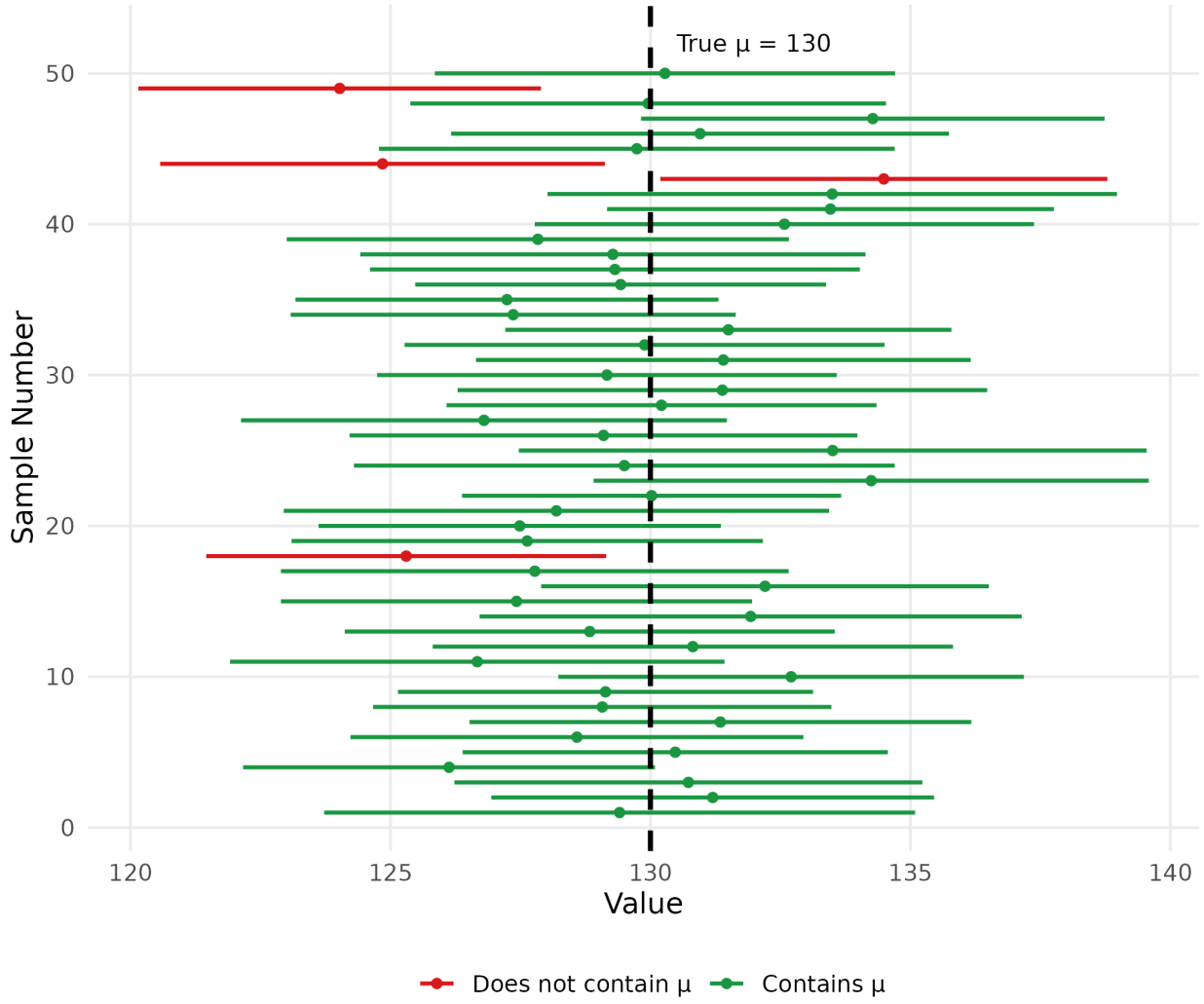


Figure 2: Simulation: 50 confidence intervals from repeated samples. Green intervals contain the true mean (= 130); red ones do not.

$df = n - 1 = 35$. From the t-table at left cumulative probability $1 - \alpha/2 = 0.975$: $t_{0.975,\,35} = 2.030$
(df = 35 is in the t-table. Since n = 36 is fairly large, the value is already close to $z_{0.975} = 1.96$.)
Step 3 — Confidence Interval:

$$105.4 \pm 2.030 \times 3.10 = 105.4 \pm 6.29$$

95% CI: (99.1, 111.7) mg/dL

Interpretation: We are 95% confident that the true mean fasting blood glucose of adults in this population is between **99.1 and 111.7 mg/dL**.

Example 2: Effect of Sample Size on CI Width

Using the same data as Example 1 ($\bar{x} = 105.4$, $s = 18.6$), compare the 95% CI when $n = 36$, $n = 100$, and $n = 400$.

Solution

Table 3: Effect of Sample Size on CI Width (xbar = 105.4, s = 18.6, 95% CI)

n	SE	t(0.975, n-1)	Margin of Error	Lower Bound	Upper Bound	CI Width
36	3.10	2.030	6.29	99.1	111.7	**12.59**
100	1.86	1.984	3.69	101.7	109.1	**7.38**
400	0.93	1.966	1.83	103.6	107.2	**3.66**

The first row is exactly the CI from Example 1.
Conclusion: Quadrupling the sample size (from 100 to 400) roughly **halves** the CI width — because SE depends on $\sqrt{n}$, not n directly (the small extra change comes from the t value getting closer to 1.96). To halve the margin of error, you need about **four times** the sample size.

Confidence Interval for a Proportion

When the outcome of interest is a **proportion** (e.g., prevalence of a disease, percentage of patients who respond to treatment), we use:

$$\hat{p} \pm z_{1-\alpha/2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Where $\hat{p} = \dfrac{x}{n}$ is the sample proportion (x = number with the characteristic, n = total sample size).

When Is This Formula Valid?

This formula requires a **sufficiently large sample**. A common rule of thumb is:

$$n\hat{p} \geq 5 \quad \text{and} \quad n(1-\hat{p}) \geq 5$$

If this condition is not met, special methods are needed (not covered in this course).

Example 3: 95% CI for a Proportion

In a survey of **200 adults**, **68 were found to have hypertension**.

Construct a **95% confidence interval** for the true prevalence of hypertension in this population.

Solution

Given: $n = 200$, $x = 68$
Step 1 — Sample proportion:

$$\hat{p} = \frac{68}{200} = 0.34 \quad (34\%)$$

Step 2 — Check validity:

$$n\hat{p} = 200 \times 0.34 = 68 \geq 5 \checkmark$$

$$n(1-\hat{p}) = 200 \times 0.66 = 132 \geq 5 \checkmark$$

Step 3 — Standard Error:

$$SE = \sqrt{\frac{0.34 \times 0.66}{200}} = \sqrt{\frac{0.2244}{200}} = \sqrt{0.001122} = 0.0335$$

Step 4 — Confidence Interval (using $z_{0.975} = 1.96$):

$$0.34 \pm 1.96 \times 0.0335 = 0.34 \pm 0.066$$

$$\textbf{95\% CI: } (0.274,\ 0.406) \quad \text{or} \quad (27.4\%,\ 40.6\%)$$

Interpretation: We are 95% confident that the true prevalence of hypertension in this population is between **27.4% and 40.6%**.

Sample Size for Estimating a Mean

Before a study starts, researchers must decide **how many participants** to recruit. Too few, and the CI will be too wide to be useful; too many, and time, money and patients' effort are wasted.

Where the Formula Comes From

The **margin of error** E is the "±" part of the CI:

$$E = z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

If we decide in advance how small we want E to be, we can solve this equation for n:

$$\sqrt{n} = \frac{z_{1-\alpha/2} \cdot \sigma}{E} \qquad \Rightarrow \qquad n = \left(\frac{z_{1-\alpha/2} \cdot \sigma}{E} \right)^2$$

Three Rules for Sample Size

1. **We use z, not t.** The t value depends on n, which we do not know yet.
2. **We need a value for** σ before the study — usually from a pilot study or a previous publication.
3. **Always round UP** to the next whole number, even if the decimal is small (e.g., 96.04 becomes 97). Rounding down would give a margin of error slightly *larger* than we wanted.

Worked Example

A researcher wants to estimate the mean systolic blood pressure of elderly patients in a district, with a **95% CI** and a margin of error of **no more than 3 mmHg**. A previous study reported **SD = 15 mmHg**.

$$n = \left(\frac{1.96 \times 15}{3} \right)^2 = (9.8)^2 = 96.04 \ \rightarrow \ \textbf{97 patients}$$

Notice the effect of each part of the formula:

- A **smaller** E (more precision) needs a **larger** n — halving E multiplies n by 4.
- A **larger** σ (more variable patients) needs a **larger** n.
- A **higher** confidence level (99% uses 2.576) needs a **larger** n.

Key Takeaways

Chapter 4 Summary

1. A **point estimate** gives a single best-guess value for a population parameter; an **interval estimate** gives a range of plausible values.
2. A **confidence interval (CI)** = Point estimate ± Margin of error. The margin of error depends on the confidence level, the SE, and the critical value.
3. Use the **Z-interval** only when σ is known. When σ is unknown (almost always), use the **t-interval** with $df = n - 1$ — this is what SPSS does. For large n, the t and Z values are very close. The **Central Limit Theorem** is why these intervals work for large samples even when the data are skewed.
4. The **95% CI** is the standard in health research, using $z_{0.975} = 1.96$.
5. A wider CI means less precision; a larger sample size n always produces a narrower (more precise) CI.
6. **Sample size for a mean:** $n = \left(z_{1-\alpha/2}\,\sigma/E\right)^2$, always rounded **up**.
7. The correct interpretation: *"We are 95% confident the true parameter lies within this interval"* — not a probability statement about a fixed parameter.

Formula Summary

Table 1: Formulas in this chapter

Quantity	Formula	Notes
SE of a mean	$SE_{\bar{x}} = \dfrac{s}{\sqrt{n}}$	Gets smaller as n increases
SE of a proportion	$SE_{\hat{p}} = \sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$	$\hat{p} = x/n$

Quantity	Formula	Notes
CI for a mean, σ known	$\bar{x} \pm z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$	Rare in practice
CI for a mean, σ unknown	$\bar{x} \pm t_{1-\alpha/2,\, n-1} \cdot \frac{s}{\sqrt{n}}$	$df = n - 1$; what SPSS uses
CI for a proportion	$\hat{p} \pm z_{1-\alpha/2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	Needs $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$
Sample size for a mean	$n = \left(\frac{z_{1-\alpha/2} \cdot \sigma}{E}\right)^2$	Always round **up**
Critical values of Z	$z_{0.95} = 1.645$, $z_{0.975} = 1.96$, $z_{0.995} = 2.576$	90%, 95%, 99% CI

Exercises

Exercise 1

A random sample of **25 patients** with chronic kidney disease had their serum creatinine levels measured:

$$\bar{x} = 3.82 \text{ mg/dL}, \qquad s = 1.14 \text{ mg/dL}$$

(a) Calculate the standard error of the mean.
(b) Find the appropriate critical value for a 95% CI.
(c) Construct the **95% confidence interval** for the population mean serum creatinine level.
(d) Interpret the confidence interval in plain language.

Exercise 2

A hospital records that **45 out of 180 patients** discharged last month were readmitted within 30 days.
(a) Calculate the sample proportion of 30-day readmissions.
(b) Check whether the large-sample conditions are satisfied.
(c) Construct a **95% confidence interval** for the true 30-day readmission rate.
(d) The hospital's target readmission rate is 20%. Based on the confidence interval, is there evidence that the true rate differs from this target? Explain.

Exercise 3

A researcher wants to estimate the mean diastolic blood pressure (DBP) of adults in a city. From a pilot study, the SD is estimated to be **12 mmHg**.
The researcher wants a **95% CI with a margin of error no greater than 2 mmHg**.
Using the formula:

$$n = \left(\frac{z_{1-\alpha/2} \cdot \sigma}{E} \right)^2$$

where E is the desired margin of error, calculate the **minimum sample size** required.
Remember to round **up**.

End of Chapter 4

Coming Up Next

Chapter 5: Statistical Testing — Instead of estimating a parameter, we will test specific claims about populations using **hypothesis testing** — one of the most powerful tools in health research.

Statistical Testing

Introduction

Why Does This Matter?

In health research, we constantly ask questions that require a decision:

- Does this new drug lower blood pressure *more* than the standard treatment?
- Is there a relationship between smoking and lung disease?
- Do recovery times differ between two surgical techniques?

We cannot answer these questions simply by looking at sample means or proportions — random chance alone can create apparent differences between groups. **Statistical testing** gives us a formal, objective framework for deciding whether an observed difference is likely to be real or simply due to chance.
This is the most widely used tool in health research, and understanding it is essential for reading and interpreting scientific literature.

Concept of Hypothesis Testing

The Logic of Hypothesis Testing

Statistical testing works by assuming the opposite of what we want to prove, then asking: *"If nothing were really happening, how surprising would our data be?"*

If the data would be very surprising under the assumption that nothing is happening, we conclude that something probably *is* happening.

The Two Hypotheses

Every hypothesis test involves two competing statements:

Null Hypothesis (H_0)

The **null hypothesis** states that there is **no effect, no difference, or no relationship** in the population. It is the assumption we start with and attempt to disprove.

> *Examples:* - The mean blood pressure is 120 mmHg. ($H_0 : \mu = 120$) - There is no difference in recovery time between the two groups. ($H_0 : \mu_1 = \mu_2$) - There is no association between smoking status and disease. ($H_0 :$ no association)

Alternative Hypothesis (H_1 or H_a)

The **alternative hypothesis** states that there **is** an effect, difference, or relationship. This is usually what the researcher believes or wants to demonstrate.

> *Examples:* - The mean blood pressure is not 120 mmHg. ($H_1 : \mu \neq 120$) - Recovery time differs between the two groups. ($H_1 : \mu_1 \neq \mu_2$) - There is an association between smoking and disease. ($H_1 :$ association exists)

One-Tailed vs. Two-Tailed Tests

The alternative hypothesis can be **directional** or **non-directional**:

Table 1: One-tailed vs. Two-tailed Tests

Test Type	H1	Use When
Two-tailed	mu != mu0 (different from)	Either direction (most common)
One-tailed (right)	mu > mu0 (greater than)	Only an increase is of interest
One-tailed (left)	mu < mu0 (less than)	Only a decrease is of interest

Practical Advice

In most health research, use a **two-tailed test** unless you have a strong, pre-specified reason to test in only one direction. Two-tailed tests are more conservative and more widely accepted. SPSS reports two-tailed p-values ("Sig. (2-tailed)"), and **all tests in this course are two-tailed**.
After a significant two-tailed test, we still say **which direction** the difference goes by looking at the sample means (or mean ranks).

The Test Statistic

A **test statistic** is a single number calculated from the sample that measures **how far the data are from what H_0 predicts**. For the t-tests it has the same form as the Z-score in Chapter 3:

$$\text{test statistic} = \frac{\text{what we observed} - \text{what } H_0 \text{ predicts}}{\text{standard error}}$$

- A test statistic **close to 0** means the data agree with H_0.
- A test statistic **far from 0** means the data are surprising if H_0 is true.

Each test in this chapter has its own test statistic (t, χ^2, U, W) and its own formula, but the idea is the same: the further the data are from what H_0 predicts, the more extreme the test statistic.

We write the value **calculated from the sample** with the subscript "cal": t_{cal}, z_{cal} or χ^2_{cal}. This keeps it apart from the **critical value** read from a table (such as $t_{0.975,\,df}$), and we compare $|t_{cal}|$ with that critical value.

The p-value

The **p-value** is the probability of obtaining results *at least as extreme as those observed*, assuming the null hypothesis is true.

$$p\text{-value} = P(\text{observed result or more extreme} \mid H_0 \text{ is true})$$

- A **small p-value** means the observed data would be very unlikely if H_0 were true $\rightarrow$ **evidence against** H_0
- A **large p-value** means the observed data are consistent with $H_0 \rightarrow$ **insufficient evidence to reject** H_0

The Significance Level ( )

The **significance level** α is the threshold we set in advance for deciding whether the p-value is "small enough" to reject H_0.

The most common choice in health research is $\alpha = 0.05$.

Decision Rule	
If p-value $\leq \alpha$	**Reject** H_0 — result is statistically significant
If p-value $> \alpha$	**Fail to reject** H_0 — insufficient evidence

Common Misinterpretations

"Failing to reject H_0" is NOT the same as proving H_0 is true. It simply means the data do not provide strong enough evidence against it.
"Statistically significant" does NOT necessarily mean clinically important. With a very large sample, even a trivially small difference can become statistically significant. Always consider the practical meaning of results.

Two Ways to Make the Decision: Critical Value or p-value

There are two equivalent ways to decide whether to reject H_0. They **always give the same decision**.

1. Critical value approach (calculating by hand). The **critical value** is the cut-off from a statistical table (Z, t or χ^2) that marks the most extreme α of the distribution. We use the same left-cumulative notation as in Chapter 4. For a two-tailed test at $\alpha = 0.05$ the cut-off is at $1 - \alpha/2 = 0.975$, for example $t_{0.975,\,9} = 2.262$.

- If |test statistic| > critical value (e.g. $|t_{cal}| > t_{0.975,\,df}$) → **reject** H_0
- If |test statistic| ≤ critical value → **do not reject** H_0

(The rank tests later in this chapter also use this rule: we convert their test statistic to a Z-score first.)

2. p-value approach (reading SPSS output). An exact p-value is hard to find by hand, but SPSS calculates it for us (the column **Sig.**). We compare it with α as in the table above.

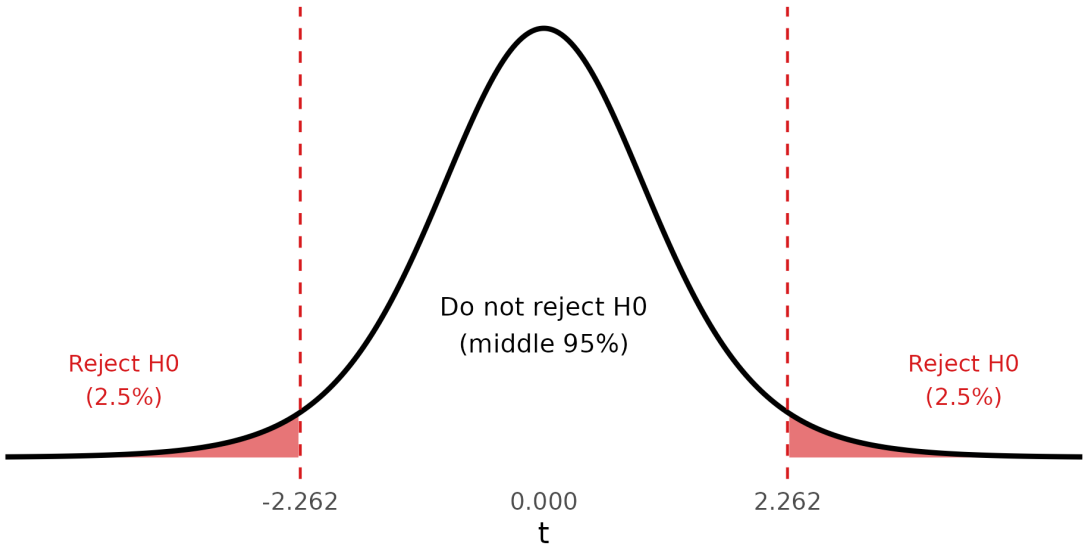


Figure 1: Two-tailed test at alpha = 0.05 with df = 9: reject H0 when the test statistic falls in a red tail beyond the critical value 2.262

In this book, worked examples use the **critical value** for the hand calculation and then show the **p-value** that SPSS would report, so you can see that both lead to the same conclusion. In the lab, you will use the p-value.

Type I and Type II Errors

When making a decision about H_0, two types of errors are possible:

Table 2: Type I and Type II Errors in Hypothesis Testing

Our Decision	H0 is TRUE	H0 is FALSE
Reject H0	Type I Error (alpha) - False Positive	Correct Decision (Power = 1 - beta)
Fail to reject H0	Correct Decision	Type II Error (beta) - False Negative

- **Type I error ( ):** Concluding there is an effect when there is none (*false positive*). Controlled by choosing (usually 0.05).
- **Type II error ( ):** Missing a real effect (*false negative*). The probability of correctly detecting a real effect is called **power** $= 1 - \beta$.

General Steps in Hypothesis Testing

The 5-Step Framework

Step 1 — State the hypotheses: Write H_0 and H_1 clearly.
Step 2 — Choose the significance level: Usually $\alpha = 0.05$.
Step 3 — Select and compute the test statistic: Choose the test that matches the data type and study design, and calculate its test statistic.
Step 4 — Compare: By hand, compare the test statistic with the **critical value** from the table; with SPSS, read the **p-value** (Sig.).
Step 5 — Make a decision and interpret: Reject or do not reject H_0, then state your conclusion in the context of the problem — including the **direction** of any difference.

Statistical Testing on Categorical Data

Chi-Square Test for Two Independent Groups

When to Use

The **chi-square test of independence** is used when:

- The outcome variable is **categorical (nominal)**
- We are comparing **two or more independent groups**
- We want to test whether there is an **association** between two categorical variables

The Contingency Table

Data are arranged in a **contingency table** (also called a cross-tabulation):

	Outcome: Yes	**Outcome: No**	**Row Total**
Group 1	a	b	$a+b$
Group 2	c	d	$c+d$
Column Total	$a+c$	$b+d$	n

The Test Statistic

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where:

- O = observed frequency (actual count in each cell)
- E = expected frequency = $\dfrac{\text{Row total} \times \text{Column total}}{n}$

The sum is taken over **all cells** in the table. The expected frequency is the count we would expect in a cell **if H_0 were true** (no association), so a large χ^2 means the observed counts are far from what H_0 predicts.

The test statistic follows a **chi-square distribution** with $df = (\text{rows} - 1) \times (\text{columns} - 1)$.

For a 2×2 table: $df = (2-1)(2-1) = 1$.

Because χ^2 is a sum of squares, it is never negative, and only **large** values are evidence against H_0. The critical value therefore uses the left cumulative probability $1 - \alpha$: at $\alpha = 0.05$ and $df = 1$, $\chi^2_{0.95,\,1} = 3.841$.

Condition for Valid Chi-Square Test

All **expected frequencies** must be ≥ 5. If this condition is violated, use **Fisher's exact test** instead (available in SPSS).

Chi-Square Critical Values

Table 1: Chi-Square Critical Values at alpha = 0.05 (left cumulative probability 0.95)

Degrees of Freedom (df)	chi2(0.95, df)
1	3.841
2	5.991
3	7.815
4	9.488
5	11.070
6	12.592
7	14.067
8	15.507
9	16.919
10	18.307

Reading the SPSS Output (Crosstabs)

SPSS prints several rows in the **Chi-Square Tests** table. In this course:

1. First read the **footnote** under the table, e.g. *"0 cells (0.0%) have expected count less than 5."*
2. If **no** cell has an expected count below 5 → read the **Pearson Chi-Square** row (Value, df, Asymp. Sig.).
3. If **any** cell has an expected count below 5 → read the **Fisher's Exact Test** row instead (Exact Sig. 2-sided).

The other rows (Continuity Correction, Likelihood Ratio, Linear-by-Linear Association) are not used in this course.

Example 1: Chi-Square Test

A cross-sectional study investigates whether **smoking status** is associated with **hypertension** in a sample of 200 adults.

	Hypertension	No Hypertension	Total
Smoker	45	55	100
Non-smoker	25	75	100
Total	**70**	**130**	**200**

Test at $\alpha = 0.05$ whether smoking is associated with hypertension.

Solution

Step 1 — Hypotheses:
H_0: Smoking status and hypertension are independent (no association).
H_1: Smoking status and hypertension are associated.
Step 2 — $\alpha = 0.05$
Step 3 — Expected frequencies and test statistic:

$$E = \frac{\text{Row total} \times \text{Column total}}{n}$$

Table 2: Observed (O) and Expected (E) Frequencies

Group	O (Hyp.)	E (Hyp.)	O (No Hyp.)	E (No Hyp.)
Smoker	45	35	55	65
Non-smoker	25	35	75	65

All expected frequencies are ≥ 5, so the chi-square test is valid.

$$\chi^2_{cal} = \frac{(45-35)^2}{35} + \frac{(55-65)^2}{65} + \frac{(25-35)^2}{35} + \frac{(75-65)^2}{65}$$

$$= \frac{100}{35} + \frac{100}{65} + \frac{100}{35} + \frac{100}{65} = 2.857 + 1.538 + 2.857 + 1.538 = \mathbf{8.791}$$

$df = (2-1)(2-1) = 1$

Step 4 — Compare:

Critical value from the chi-square table: $\chi^2_{0.95,\,1} = 3.841$. Since $\chi^2_{cal} = 8.791 > 3.841$, the test statistic falls in the rejection region.

SPSS (Pearson Chi-Square row) would report $p = 0.003$, which is < 0.05 — the same decision.

Step 5 — Decision and interpretation:

We **reject** H_0. There is a statistically significant association between smoking status and hypertension ($\chi^2 = 8.79$, $df = 1$, $p = 0.003$). Smokers have a higher prevalence of hypertension (45%) than non-smokers (25%).

*Because this is a cross-sectional study, the result shows an **association**, not that smoking **causes** hypertension.*

Statistical Testing on Quantitative Data

The two t-tests in this section compare **means**. Like the confidence intervals in Chapter 4, they rely on the data (or the sample means) being approximately **normal**.

Checking the Normality Assumption

Before using a t-test, check whether the outcome is approximately normal:

- **Graphs:** a histogram should look roughly bell-shaped, and the points on a **normal Q–Q plot** should lie close to the straight line.
- **Tests:** SPSS *Explore* reports the **Kolmogorov–Smirnov** and **Shapiro–Wilk** tests (you practise these in the lab). Their hypotheses are

  H_0: the data come from a normal distribution

  H_1: the data do not come from a normal distribution

  so **Sig. $>$ 0.05** means *no evidence against normality* $\rightarrow$ a t-test is appropriate; **Sig. $\leq$ 0.05** means the data are not normal $\rightarrow$ consider a rank test (later in this chapter).

What to check depends on the design:

Test	Check normality of
Independent samples t-test	The outcome **in each group separately**
Paired samples t-test	The **differences** d_i within pairs

With large samples ($n \geq 30$ per group) the **Central Limit Theorem** (Chapter 4) makes the t-test robust even if the data are somewhat skewed.

Independent Samples t-Test

When to Use

The **independent samples t-test** is used when:

- The outcome variable is **continuous (numerical)**
- We are comparing the **means of two independent groups**
- The data are approximately **normally distributed** in each group (or sample sizes are large enough, $n \geq 30$ per group)

Hypotheses

$$H_0 : \mu_1 = \mu_2 \quad \text{(no difference in population means)}$$
$$H_1 : \mu_1 \neq \mu_2 \quad \text{(the population means differ)}$$

The Test Statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE_{\bar{x}_1 - \bar{x}_2}}$$

The standard error of the difference depends on whether the two groups have **equal or unequal variances**.

Equal variances assumed (pooled):

$$SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, \qquad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Equal variances not assumed (Welch's):

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

The test statistic follows a **t-distribution** with $df = n_1 + n_2 - 2$ (equal variances) or an approximated df (Welch's, calculated by SPSS).

Which Version? Levene's Test

SPSS decides between the two versions with **Levene's test for equality of variances**, printed in the first columns of the *Independent Samples Test* table:

$H_0 : \sigma_1^2 = \sigma_2^2$ (the two variances are equal)

$H_1 : \sigma_1^2 \neq \sigma_2^2$ (the variances differ)

Reading the Independent Samples Test Table — Two Steps

Step 1 — Levene's test (Sig. in the Levene columns):

- Sig. $>$ **0.05** $\rightarrow$ no evidence that the variances differ $\rightarrow$ read the **top row**, *"Equal variances assumed"*.
- Sig. $\leq$ **0.05** $\rightarrow$ the variances differ $\rightarrow$ read the **bottom row**, *"Equal variances not assumed"*.

Step 2 — the t-test (Sig. (2-tailed) in the row you chose): compare with α to decide whether the **means** differ.
Do not confuse the two Sig. values: the first is about **variances**, the second is about **means**.

When you calculate by hand in this course, the question will either give you the Levene's test result or state whether the variances can be assumed equal. You do not need to decide this yourself from the SDs.

Example 2: Independent Samples t-Test

A clinical trial compares the mean reduction in systolic blood pressure (mmHg) between patients receiving a **new drug** vs. a **placebo** after 8 weeks.

Group	n	Mean reduction	SD
New drug	40	12.4	6.8
Placebo	40	7.1	7.2

Assume the two population variances are **equal**. Test at $\alpha = 0.05$ whether the mean reduction in SBP differs between the new drug and placebo.

Solution

Step 1 — Hypotheses:
$H_0 : \mu_1 = \mu_2$ (no difference in mean reduction)
$H_1 : \mu_1 \neq \mu_2$ (two-tailed; difference exists)
Step 2 — $\alpha = 0.05$
Step 3 — Pooled standard error. The question states that the variances are equal, so we use the pooled (equal variances) formula.

$$s_p = \sqrt{\frac{(40-1)(6.8)^2 + (40-1)(7.2)^2}{40+40-2}} = \sqrt{\frac{39(46.24) + 39(51.84)}{78}}$$
$$= \sqrt{\frac{1803.36 + 2021.76}{78}} = \sqrt{49.041} = 7.003$$

$$SE = 7.003 \times \sqrt{\frac{1}{40} + \frac{1}{40}} = 7.003 \times \sqrt{0.05} = 7.003 \times 0.2236 = 1.566$$

$$t_{cal} = \frac{12.4 - 7.1}{1.566} = \frac{5.3}{1.566} = \mathbf{3.384}$$

$df = 40 + 40 - 2 = 78$
Step 4 — Compare:
$df = 78$ is not listed in the t-table, so we use the next smaller df, 70: $t_{0.975,\,70} = 1.994$ (the exact value for $df = 78$ is 1.991).
Since $|t_{cal}| = 3.384 > 1.994$, the test statistic falls in the rejection region. SPSS would report $p = 0.001$, which is < 0.05.
Step 5 — Decision and interpretation:
We **reject** H_0. The mean reduction in systolic blood pressure differs significantly between the two groups ($t = 3.38$, $df = 78$, $p = 0.001$). The new drug group had the **larger** mean reduction: 12.4 vs. 7.1 mmHg, a difference of 5.3 mmHg (95% CI: 2.2 to 8.4 mmHg).

Paired Samples t-Test

When to Use

The **paired samples t-test** is used when:

- The outcome variable is **continuous (numerical)**
- Observations come in **natural pairs** — the same subject measured twice (e.g., before and after treatment), or matched pairs of subjects
- We want to test whether the **mean difference** within pairs is zero

- The **differences** are approximately normally distributed

Hypotheses

$$H_0 : \mu_d = 0 \quad \text{(mean difference is zero)}$$
$$H_1 : \mu_d \neq 0 \quad \text{(mean difference is not zero)}$$

Where μ_d is the population mean of the within-pair differences.

The Test Statistic

For each pair i, compute the difference: $d_i = x_{1i} - x_{2i}$

Then:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Where:

- $\bar{d}$ = mean of the differences
- s_d = standard deviation of the differences
- n = number of pairs

The test statistic follows a **t-distribution** with $df = n - 1$.

> Why Not Use the Independent t-Test for Paired Data?
>
> Each patient is compared **with themselves**, so differences between patients (age, weight, baseline health) cancel out. Working with the differences d_i turns two columns of data into **one**, and the test becomes a one-sample test of whether the mean difference is 0.

Example 3: Paired Samples t-Test

A study measures the **diastolic blood pressure (DBP)** of 10 patients **before** and **after** 4 weeks of a lifestyle intervention.

Table 1: DBP Before and After Lifestyle Intervention (n = 10 patients)

Patient	DBP Before (mmHg)	DBP After (mmHg)	Difference d = Before - After
Patient 1	92	85	**7**
Patient 2	88	82	**6**
Patient 3	95	89	**6**
Patient 4	100	91	**9**
Patient 5	85	80	**5**
Patient 6	97	88	**9**
Patient 7	90	84	**6**
Patient 8	94	87	**7**
Patient 9	88	83	**5**
Patient 10	96	88	**8**

Test at $\alpha = 0.05$ whether the mean DBP changed after the intervention. Assume the differences are approximately normally distributed.

Solution

Step 1 — Hypotheses:
$H_0 : \mu_d = 0$ (no mean change in DBP)
$H_1 : \mu_d \neq 0$ (mean DBP has changed)
Step 2 — $\alpha = 0.05$
Step 3 — Compute $\bar{d}$, s_d and the test statistic:

$$\bar{d} = \frac{68}{10} = 6.8 \text{ mmHg}$$

$$s_d = \sqrt{\frac{\sum(d_i - \bar{d})^2}{n-1}} = 1.476$$

$$t_{cal} = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{6.8}{1.476/\sqrt{10}} = \frac{6.8}{0.467} = 14.571$$

$df = n - 1 = 9$
Step 4 — Compare:
$t_{0.975,\,9} = 2.262$. Since $|t_{cal}| = 14.571 > 2.262$, the test statistic falls in the rejection region. SPSS would report $p < 0.001$.
Step 5 — Decision and interpretation:
We **reject** H_0. Mean diastolic blood pressure changed significantly after the intervention ($t = 14.57$, $df = 9$, $p < 0.001$). Because $\bar{d}$ = Before − After is positive, DBP **decreased**, by **6.8 mmHg** on average.

Statistical Testing on Rank Data

The t-tests need **numerical** data that are approximately **normal**. When the outcome is **ordinal** (such as a pain score) or the normality check fails, we use **rank tests** instead. Rank tests replace each value by its **rank** (1st, 2nd, 3rd, …) and do not assume any particular distribution — they are called **non-parametric** tests.

Each t-test has a rank-test partner:

Design	Numerical, normal	Ordinal, or not normal
Two **independent** groups	Independent samples t-test	**Mann-Whitney U test**
Two **related** (paired) measurements	Paired samples t-test	**Wilcoxon signed-ranks test**

Rank Tests Use the Z-Table You Already Know

The rank tests have their own statistics (U and W), but we do not need a new table. We convert U or W to a **Z-score** — exactly what SPSS does — and then use the same rule as in Chapter 3 and Chapter 4:

$$\text{reject } H_0 \text{ if } |z| > z_{0.975} = 1.96 \quad (\alpha = 0.05,\ \text{two-tailed})$$

Mann-Whitney U Test for Two Independent Groups

When to Use

The **Mann-Whitney U test** (also called the Wilcoxon rank-sum test) is used when:

- The outcome variable is **ordinal**, or
- The outcome is **continuous but not normally distributed** (especially with small samples or obvious skewness)
- We are comparing **two independent groups**

It is the non-parametric alternative to the **independent samples t-test**.

Hypotheses

H_0: the outcome values in the two groups are equal (the two groups come from the same distribution)

H_1: the outcome values in the two groups are different

How It Works

Instead of comparing means, the Mann-Whitney test compares the **ranks** of observations between the two groups.

Steps:

1. Combine all observations from both groups and **rank them** from smallest (rank 1) to largest.
2. If there are **tied values**, assign the average rank to each tied observation.
3. Calculate the **sum of ranks** for each group: W_1 and W_2.
4. Calculate the test statistic U from the rank sums (formula below).
5. Convert U to a **Z-score** (below) and compare $|z|$ with 1.96. SPSS reports this Z-score and its p-value.

> Intuition Behind the Test
>
> If the two groups come from the same distribution, their ranks should be **randomly mixed** between the groups — neither group should consistently have higher or lower ranks. A significant result means one group consistently has higher ranks (i.e., higher values) than the other.

The U Statistic

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - W_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - W_2$$

$$U = \min(U_1, U_2)$$

Where n_1, n_2 are the sample sizes and W_1, W_2 are the rank sums for groups 1 and 2 respectively. As a check, $U_1 + U_2 = n_1 n_2$.

A **small** U means the two groups hardly overlap — evidence against H_0.

Converting U to a Z-Score

If H_0 is true, U is centred at $n_1 n_2/2$ (half of the largest possible value). We standardise it like any Z-score — *(value − mean) / SD*:

$$\mu_U = \frac{n_1 n_2}{2}, \qquad \sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}, \qquad z = \frac{U - \mu_U}{\sigma_U}$$

- **Decision:** reject H_0 if $|z| > 1.96$.
- **p-value (optional):** $p = 2 \times P(Z < -|z|) = 2\,[\,1 - P(Z < |z|)\,]$, read from the Z-table.

Because $U = \min(U_1, U_2)$ is the smaller value, z is always zero or negative. SPSS prints the same Z-score in its *Test Statistics* table.

Why Your Z May Differ Slightly From SPSS

When there are **tied values**, SPSS adjusts σ_U for the ties, so its Z-score can differ from the hand calculation in the second decimal place. The decision is the same in practice. The Z-approximation is most accurate when each group has about 10 or more observations; SPSS uses the same approximation for its *Asymp. Sig.* value.

Example 4: Mann-Whitney U Test

Pain score (0–10, **ordinal**) was recorded 24 hours after surgery in **12 patients**: 6 received the standard analgesic regimen and 6 received a new regimen. The two groups are different patients, so they are **independent**.

Regimen	Pain scores
Standard	6, 7, 5, 8, 6, 7
New	3, 4, 5, 2, 4, 6

Test at $\alpha = 0.05$ whether pain scores differ between the two regimens.

Solution

Hypotheses: H_0: pain scores in the two groups are equal; H_1: pain scores in the two groups are different. $\alpha = 0.05$.

Table 1: Pain scores and their ranks (n1 = n2 = 6)

Group	Pain score	Rank (combined)

New	2	1.0
New	3	2.0
New	4	3.5
New	4	3.5
Standard	5	5.5
New	5	5.5
Standard	6	8.0
Standard	6	8.0
New	6	8.0
Standard	7	10.5
Standard	7	10.5
Standard	8	12.0

Step 1 — Rank all 12 scores together. Ties share the average rank: the two 4s occupy ranks 3 and 4, so each gets 3.5; the three 6s occupy ranks 7, 8 and 9, so each gets 8.

Step 2 — Sum the ranks in each group:

$$W_{\text{Standard}} = 8 + 10.5 + 5.5 + 12 + 8 + 10.5 = 54.5$$

$$W_{\text{New}} = 2 + 3.5 + 5.5 + 1 + 3.5 + 8 = 23.5$$

(Check: the total of ranks 1 to 12 is $12 \times 13/2 = 78 = 54.5 + 23.5$.)

Step 3 — Calculate U with $n_1 = n_2 = 6$:

$$U_1 = 36 + \frac{6 \times 7}{2} - 54.5 = 36 + 21 - 54.5 = 2.5$$

$$U_2 = 36 + 21 - 23.5 = 33.5$$

$$U = \min(2.5,\ 33.5) = 2.5$$

Step 4 — Convert U to a Z-score with $n_1 = n_2 = 6$:

$$\mu_U = \frac{6 \times 6}{2} = 18, \qquad \sigma_U = \sqrt{\frac{6 \times 6 \times 13}{12}} = \sqrt{39} = 6.245$$

$$z_{cal} = \frac{2.5 - 18}{6.245} = -2.48$$

Since $|z_{cal}| = 2.48 > 1.96$, the result is significant. From the Z-table, $p = 2 \times (1 - 0.9934) = 0.013$.

SPSS reports $Z = -2.513$ and p (Asymp. Sig. 2-tailed) $= 0.012$ — the same decision (the small difference is the tie adjustment).

Step 5 — Decision. We **reject** H_0. Pain scores differ significantly between the two regimens. The mean rank of the new regimen ($23.5/6 = 3.9$) is lower than that of the standard regimen ($54.5/6 = 9.1$), so patients on the **new regimen reported less pain**.

Wilcoxon Signed-Ranks Test for Two Related Groups

When to Use

The **Wilcoxon signed-ranks test** (also called the Wilcoxon matched-pairs test) is the non-parametric alternative to the **paired samples t-test**. It is used when:

- The outcome variable is **ordinal**, or
- The **differences** between pairs are **not normally distributed** (especially with small samples)
- The two sets of measurements are **related (paired)** — for example, the same patients measured before and after treatment, or two measurements taken on the same blood sample

> Mann-Whitney or Wilcoxon?
>
> Both tests work with ranks, so they are easy to confuse. The difference is the **study design**, exactly as with the two t-tests:
>
> - **Two independent groups** (different people in each group) → **Mann-Whitney U test**
> - **Two related groups** (the same people measured twice) → **Wilcoxon signed-ranks test**

Hypotheses

H_0: the two related measurements are equal

H_1: the two related measurements are different

How It Works

The Wilcoxon test looks at the **differences within each pair**, and then ranks the **sizes** of those differences.

Steps:

1. For each pair, calculate the difference $d_i = x_{i,1} - x_{i,2}$.
2. **Discard** any pair whose difference is zero.
3. Rank the **absolute** differences $|d_i|$ from smallest to largest, assigning average ranks to ties.
4. Attach to each rank the **sign** of its original difference.
5. Add the ranks belonging to the positive differences (W^+) and the ranks belonging to the negative differences (W^-).

The Test Statistic

$$W = \min(W^+,\ W^-)$$

If the two conditions really are the same, the positive and negative ranks should roughly balance, so W^+ and W^- should be similar. A **small** value of W means the differences fall mostly in one direction, which is evidence against H_0.

Converting W to a Z-Score

n is the number of pairs **after discarding zero differences**. If H_0 is true, W is centred at half of the total of all ranks:

$$\mu_W = \frac{n(n+1)}{4}, \qquad \sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{24}}, \qquad z = \frac{W - \mu_W}{\sigma_W}$$

As with Mann-Whitney: **reject** H_0 **if** $|z| > 1.96$, and SPSS reports this Z-score (adjusted for ties) in its *Test Statistics* table.

Reading the SPSS Output

Choose **Analyze → Nonparametric Tests → Legacy Dialogs → 2 Related Samples**, and tick **Wilcoxon** (older SPSS versions, as in the lab manual, have no *Legacy Dialogs* step). The *Ranks* table shows the sum of the negative and positive ranks, and the *Test Statistics* table reports Z and the 2-tailed significance. As always, decide by comparing the p-value with α — the sign of Z only indicates the direction of the ranks, not the importance of the result.

Example 5: Wilcoxon Signed-Ranks Test

Blood clotting time (minutes) was measured for **8 patients** using two solutions: an extract from shrimp shells ("Shrimp") and the standard solution ("Standard"). Blood was taken from each patient and divided into two parts, so the two measurements are **paired**.

Patient	1	2	3	4	5	6	7	8
Shrimp	3	5	5	4	3	7	8	7
Standard	7	6	3	8	5	9	7	9

The sample is small ($n = 8$), so we cannot rely on the normality of the differences. Test at $\alpha = 0.05$ whether the clotting times differ, using the **Wilcoxon signed-ranks test**.

Solution

Hypotheses: H_0: the clotting times with the two solutions are equal; H_1: they are different. $\alpha = 0.05$.

Table 2: Blood clotting time using two solutions (n = 8 patients), with signed ranks

Patient	Shrimp	Standard	d	abs(d)	Rank of abs(d)	Signed rank
1	3	7	-4	4	7.5	-7.5
2	5	6	-1	1	1.5	-1.5
3	5	3	2	2	4.5	4.5
4	4	8	-4	4	7.5	-7.5
5	3	5	-2	2	4.5	-4.5
6	7	9	-2	2	4.5	-4.5
7	8	7	1	1	1.5	1.5
8	7	9	-2	2	4.5	-4.5

Step 1 — Differences $d = \text{Shrimp} - \text{Standard}$: none is zero, so all 8 pairs are kept.
Step 2 — Rank $|d|$ with average ranks for ties: the two 1s share ranks 1–2 (each 1.5); the four 2s share ranks 3–6 (each 4.5); the two 4s share ranks 7–8 (each 7.5).
Step 3 — Sum the signed ranks:

$$W^+ = 4.5 \text{ (patient 3)} + 1.5 \text{ (patient 7)} = 6$$

$$W^- = 7.5 + 1.5 + 7.5 + 4.5 + 4.5 + 4.5 = 30 \quad \text{(patients } 1, 2, 4, 5, 6, 8\text{)}$$

(Check: $W^+ + W^- = 8 \times 9/2 = 36$.)

$$W = \min(6,\ 30) = 6$$

Step 4 — Convert W to a Z-score with $n = 8$ pairs:

$$\mu_W = \frac{8 \times 9}{4} = 18, \qquad \sigma_W = \sqrt{\frac{8 \times 9 \times 17}{24}} = \sqrt{51} = 7.141$$

$$z_{cal} = \frac{6 - 18}{7.141} = -1.68$$

Since $|z_{cal}| = 1.68 \leq 1.96$, the result is not significant. From the Z-table, $p = 2 \times (1 - 0.9535) = 0.093$.
SPSS reports $Z = -1.706$ and $p = 0.088$ — the same decision. *(SPSS calculates the difference as Standard − Shrimp, so its Ranks table lists the 6 and the 30 the other way round; W and the p-value are the same.)*
Step 5 — Decision. We **do not reject** H_0. There is not enough evidence to say that the clotting times produced by the two solutions differ, at the 0.05 level of significance.

Choosing the Right Test

Table 3: Guide to Choosing the Correct Statistical Test (Two Groups)

Outcome Data Type	Study Design	Statistical Test	SPSS Procedure
Categorical (Nominal)	2 independent groups	**Chi-square test**	Analyze -> Descriptive Statistics -> Crosstabs
Continuous (normal)	2 independent groups	**Independent samples t-test**	Analyze -> Compare Means -> Independent-Samples T Test
Continuous (normal)	2 related (paired) groups	**Paired samples t-test**	Analyze -> Compare Means -> Paired-Samples T Test
Ordinal or non-normal continuous	2 independent groups	**Mann-Whitney U test**	Analyze -> Nonparametric Tests -> Legacy Dialogs -> 2 Independent Samples
Ordinal or non-normal continuous	2 related (paired) groups	**Wilcoxon signed-ranks test**	Analyze -> Nonparametric Tests -> Legacy Dialogs -> 2 Related Samples

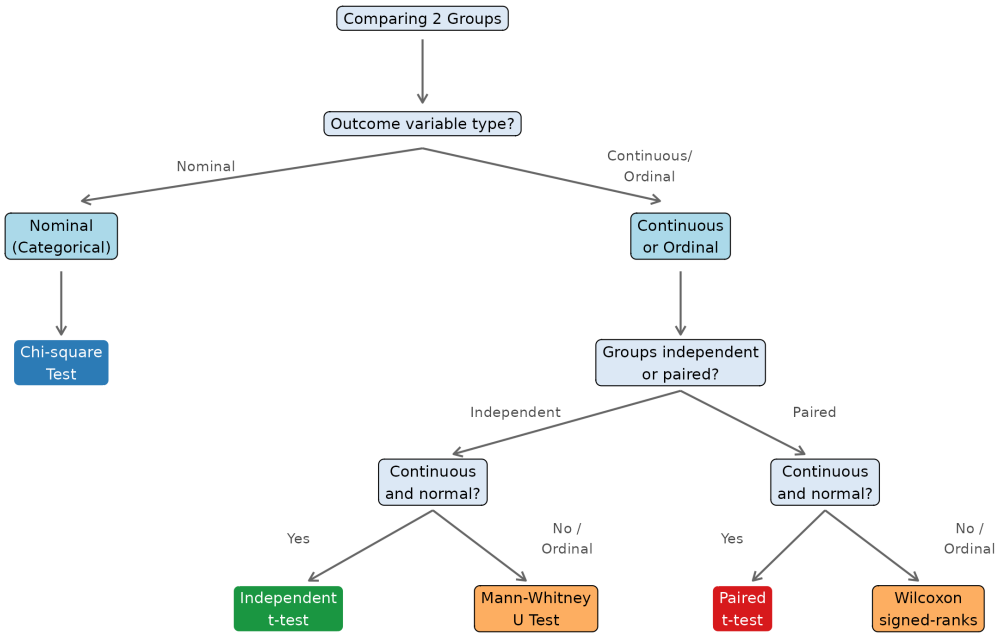


Figure 1: Decision flowchart for selecting the correct test (two-group comparisons)

Ordinal Outcomes Always Go to a Rank Test

An ordinal outcome — such as a **pain score**, a satisfaction rating or a stage of disease — is never analysed with a t-test, even if it is written as numbers. Use the Mann-Whitney U test (independent groups) or the Wilcoxon signed-ranks test (paired).

Key Takeaways

Chapter 5 Summary

1. Hypothesis testing uses sample data to decide between two competing claims: the **null hypothesis (H_0)** and the **alternative hypothesis (H_1)**.
2. The **test statistic** measures how far the data are from H_0. By hand we compare it with a **critical value**; in SPSS we compare the **p-value** with α. Both give the same decision.
3. The **p-value** is the probability of the observed data (or more extreme) if H_0 is true. A p-value $\leq \alpha$ (usually 0.05) leads us to **reject** H_0.
4. **Type I error ( ):** rejecting a true H_0 (false positive). **Type II error ( ):** failing to reject a false H_0 (false negative).
5. **Chi-square test:** for association between two categorical (nominal) variables in independent groups. Read the Pearson row, or Fisher's exact test if any expected count is below 5.
6. **Independent samples t-test:** for comparing means of two independent groups with continuous, approximately normal data. Read **Levene's test** first to choose the row.
7. **Paired samples t-test:** for comparing means of two related measurements (before/after, or matched pairs) when the differences are approximately normal.
8. **Mann-Whitney U test:** for comparing two independent groups when the outcome is ordinal or the normality assumption is not met.
9. **Wilcoxon signed-ranks test:** for comparing two **related** measurements when the outcome is ordinal or the differences are not normally distributed. It is the non-parametric alternative to the paired t-test. For both rank tests, convert U or W to a **Z-score** and **reject H_0 if** $|z| > 1.96$ — the same Z-score that SPSS reports.
10. Always choose the test based on **data type** and **study design** — using the wrong test leads to invalid conclusions.

Formula Summary

Table 1: Formulas in this chapter

Quantity	Formula	Notes
Decision with SPSS	Reject H_0 if $p \leq \alpha$	Usually $\alpha = 0.05$
Expected frequency	$E = \dfrac{\text{Row total} \times \text{Column total}}{n}$	All $E \geq 5$, otherwise Fisher's exact test
Chi-square	$\chi^2 = \sum \dfrac{(O-E)^2}{E}$	$df = (r-1)(c-1)$; reject if $\chi^2 > \chi^2_{0.95,\,df}$
Pooled SD	$s_p = \sqrt{\dfrac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$	Equal variances (Levene Sig. > 0.05)
Independent t	$t = \dfrac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$	$df = n_1 + n_2 - 2$
Welch SE	$SE = \sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$	Unequal variances; df from SPSS
Paired t	$t = \dfrac{\bar{d}}{s_d/\sqrt{n}}$	$d_i = x_{1i} - x_{2i}$; $df = n - 1$
Decision for t	Reject H_0 if $\lvert t\rvert > t_{0.975,\,df}$	Two-tailed, $\alpha = 0.05$
Mann-Whitney U	$U_1 = n_1 n_2 + \dfrac{n_1(n_1+1)}{2} - W_1$, $U = \min(U_1, U_2)$	W_1 = rank sum of group 1
U to Z	$z = \dfrac{U - n_1 n_2/2}{\sqrt{n_1 n_2 (n_1+n_2+1)/12}}$	
Wilcoxon W	$W = \min(W^+, W^-)$	Drop zero differences first
W to Z	$z = \dfrac{W - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}}$	n = number of non-zero pairs
Decision for U and W	Reject H_0 if $\lvert z\rvert > 1.96$	Same Z as SPSS (up to ties)

Exercises

Exercise 1 — Chi-Square Test

A researcher investigates whether **physical activity level** is associated with **diabetes status** in 300 adults:

	Diabetes	No Diabetes	Total
Active	18	132	150
Inactive	42	108	150
Total	**60**	**240**	**300**

(a) State H_0 and H_1.
(b) Calculate all expected frequencies and verify the chi-square condition is met.
(c) Calculate the chi-square test statistic.
(d) At $\alpha = 0.05$, what is your conclusion?
(e) In plain language, what does your result mean?

Exercise 2 — Independent Samples t-Test

A study compares the **mean hemoglobin levels (g/dL)** between male and female patients:

Group	n	$\bar{x}$	s
Male	35	14.8	1.6
Female	35	12.9	1.4

In SPSS, Levene's test for these data gives Sig. $= 0.41$.
(a) State H_0 and H_1 (two-tailed).
(b) Based on Levene's test, should you use the "Equal variances assumed" or the "Equal variances not assumed" version? Why?
(c) Calculate the pooled standard error.
(d) Calculate the t test statistic.
(e) At $\alpha = 0.05$ with appropriate df, state your conclusion.

Exercise 3 — Paired Samples t-Test

A diabetes clinic measures the **capillary blood glucose (mg/dL)** of 8 patients with type 2 diabetes **before** and **2 hours after** a structured 30-minute walking session:

Patient	1	2	3	4	5	6	7	8
Before	182	165	210	148	195	172	160	188
After	160	150	182	141	170	158	152	165

Assume the differences are approximately normally distributed.
(a) Calculate the difference d_i = Before − After for each patient.
(b) Calculate $\bar{d}$ and s_d.
(c) Calculate the paired t statistic.
(d) At $\alpha = 0.05$, did blood glucose change significantly after the walking session? In which direction? State your conclusion clearly.
(e) Which variable would you check for normality in SPSS before using this test, and which SPSS procedure would you use?

Exercise 4 — Mann-Whitney U Test

A physiotherapy team records the **time to first walk after knee surgery (hours)**, which is strongly right-skewed, for two independent groups of patients:

Group	Time to first walk (hours)
Early mobilisation programme	12, 18, 9, 30, 15
Usual care	22, 35, 28, 40, 26, 48

(a) Explain why the Mann-Whitney U test is used rather than the independent samples t-test.
(b) Rank all 11 values together and find the rank sums W_1 and W_2.
(c) Calculate U_1, U_2 and U.
(d) Convert U to a Z-score and state your conclusion at $\alpha = 0.05$. Use the Z-table to find the approximate p-value.

Exercise 5 — Wilcoxon Signed-Ranks Test

An anxiety score (0–20, **ordinal**) was recorded for 9 patients before and after a pre-operative education session:

Patient	1	2	3	4	5	6	7	8	9
Before	14	11	16	9	13	15	12	10	17
After	9	10	11	8	6	12	4	13	7

(a) Calculate $d = \text{Before} - \text{After}$ for each patient.
(b) Rank $|d|$ (use average ranks for ties) and find W^+, W^- and W.
(c) Convert W to a Z-score and state your conclusion at $\alpha = 0.05$. Did anxiety go up or down?

Exercise 6 — Choosing the Right Test

For each scenario below, state which statistical test is most appropriate and briefly explain why:
(a) Comparing the proportion of post-operative complications between patients who received standard care vs. enhanced recovery protocol.
(b) Comparing patient satisfaction scores (rated: Poor / Fair / Good / Excellent) between two hospital wards.
(c) Measuring the effect of a new diet on body weight by recording each participant's weight before and after a 12-week program (the differences are approximately normal).
(d) Comparing the mean length of hospital stay (days) between two age groups (under 60 vs. 60 and over), where length-of-stay data is strongly right-skewed.
(e) Comparing pain scores (0–10, ordinal) recorded for the **same** patients before and 24 hours after starting a new analgesic.

End of Chapter 5

Coming Up Next

Chapter 6: Introduction to Statistical Modeling — We take our first steps beyond comparing two groups, exploring how one variable can be used to explain or predict another through **linear regression**.

Introduction to Statistical Modeling

Introduction

Why Does This Matter?

So far we have described data, estimated population parameters, and tested whether two groups differ. But many real-world health questions go beyond simple comparisons:

- Does body weight *predict* blood pressure?
- As age *increases*, how does cholesterol *change*?
- Can we use a patient's BMI to *estimate* their fasting glucose level?

These questions require us to model the **relationship between two variables**. **Statistical modeling** is the process of building a mathematical equation that describes how one variable is related to another — and using that equation to understand, explain, and predict outcomes.
In this chapter we focus on the simplest and most widely used model: **simple linear regression**.

What Is a Statistical Model?

A **statistical model** is a mathematical equation that describes the relationship between:

- An **outcome variable** (also called the *dependent variable*, *response variable*, or Y) — the variable we want to explain or predict.
- One or more **predictor variables** (also called *independent variables*, *explanatory variables*, or X) — the variables we use to explain or predict Y.

In this chapter we study **simple linear regression** — one outcome Y and one predictor X, with a straight-line relationship.

Correlation: Measuring the Relationship

Before building a model, we first ask: *Is there a relationship between X and Y, and how strong is it?*

The Scatter Plot

A **scatter plot** displays pairs of values (x_i, y_i) as points on a graph. When you read one, describe four things:

- **Direction** — positive (points go up to the right) or negative.
- **Strength** — how tightly the points follow a line.
- **Form** — is the pattern **linear** (a straight line) or **curved**? Pearson's r and linear regression only describe a straight-line pattern; a clear curve can give a small r even when X and Y are closely related.
- **Outliers** — a point far from the pattern of the others. A single outlier can change r and the regression line a lot, so check it before calculating.

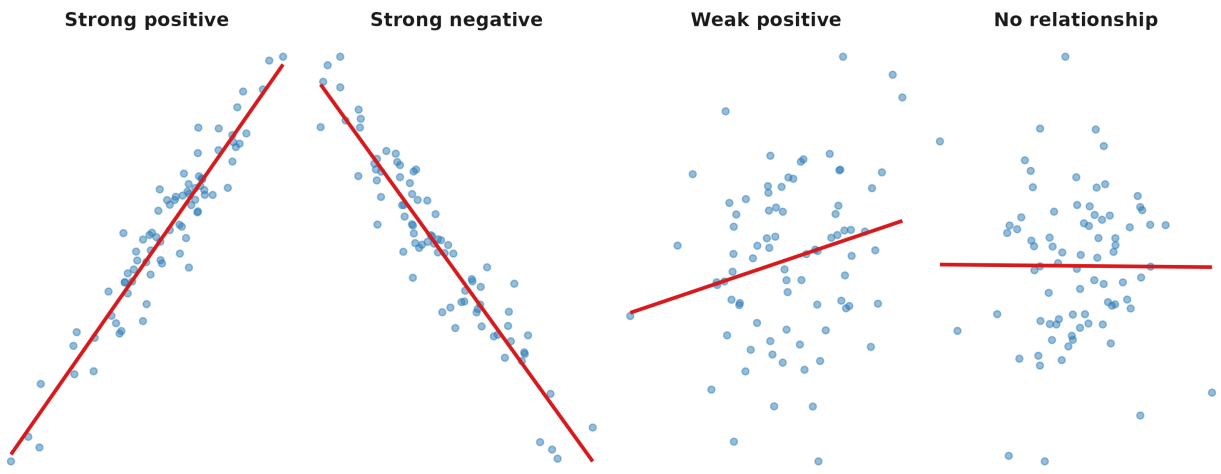


Figure 1: Four possible relationships between two variables

Pearson's Correlation Coefficient (r)

The **Pearson correlation coefficient** r quantifies the **strength** and **direction** of the *linear* relationship between two continuous variables.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \cdot \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

This **definition** shows what r measures: whether x and y tend to be above (or below) their means *together*. For calculating by hand, the same value is easier to get from five column totals, $\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$ and $\sum xy$ — no means or deviations needed:

Computational Formula for r (Use This by Hand)

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left[n\sum x^2 - \left(\sum x\right)^2\right]\left[n\sum y^2 - \left(\sum y\right)^2\right]}}$$

Note that $\sum x^2$ (square each value, then add) is **not** the same as $\left(\sum x\right)^2$ (add the values, then square the total).

Properties of r

- r always lies between $\mathbf{-1}$ **and** $\mathbf{+1}$
- $r = +1$: perfect positive linear relationship
- $r = -1$: perfect negative linear relationship
- $r = 0$: no linear relationship

Interpreting the Magnitude of r

Table 1: General guidelines for interpreting the correlation coefficient

Size of r (ignoring sign)	Strength	Note
0.00 - 0.19	Negligible	-
0.20 - 0.39	Weak	Positive or negative depending on sign
0.40 - 0.59	Moderate	Positive or negative depending on sign
0.60 - 0.79	Strong	Positive or negative depending on sign
0.80 - 1.00	Very strong	Positive or negative depending on sign

Testing Whether r Is Significant

A sample correlation can be different from 0 just by chance, especially in a small sample. We test the **population correlation** ρ ("rho"):

$$H_0 : \rho = 0 \quad \text{(no linear relationship in the population)}$$
$$H_1 : \rho \neq 0 \quad \text{(there is a linear relationship)}$$

The test statistic is

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \qquad df = n - 2$$

In SPSS (**Analyze** → **Correlate** → **Bivariate**), the **Correlations** table shows three numbers for each pair of variables:

		BMI	Glucose
BMI	Pearson Correlation	1	.612
	Sig. (2-tailed)		.000
	N	60	60

- **Pearson Correlation** = $r = 0.612$ (strong positive)
- **Sig. (2-tailed)** = the p-value. SPSS shows **.000** when $p < 0.0005$; write it as $p < 0.001$, never as $p = 0$.
- **N** = number of pairs

Since $p < 0.05$, we reject H_0: there is a significant positive linear relationship between BMI and fasting glucose.

> Correlation Is Not Causation
>
> A high correlation between X and Y does **not** mean that X *causes* Y. Both variables may be influenced by a third variable, or the relationship may be coincidental. Always interpret correlations in the context of biological or clinical plausibility.

Example 1: Calculating and Testing Pearson's r

A researcher records the **age (years)** and **systolic blood pressure (mmHg)** of 8 adults:

Table 2: Age and Systolic Blood Pressure of 8 Adults

Subject	Age (years)	SBP (mmHg)
S1	25	118
S2	32	116
S3	40	131
S4	48	128
S5	55	146
S6	62	139
S7	70	160
S8	78	157

(a) Calculate Pearson's r using the computational formula.

(b) Test at $\alpha = 0.05$ whether there is a significant linear relationship between age and systolic blood pressure.

Solution

Table 3: Column totals needed for r (the same totals are reused for the regression line in Example 2)

Subject	x	y	x^2	y^2	xy
S1	25	118	625	13,924	2,950
S2	32	116	1,024	13,456	3,712
S3	40	131	1,600	17,161	5,240
S4	48	128	2,304	16,384	6,144
S5	55	146	3,025	21,316	8,030
S6	62	139	3,844	19,321	8,618
S7	70	160	4,900	25,600	11,200
S8	78	157	6,084	24,649	12,246
Total	**410**	**1,095**	**23,406**	**151,811**	**58,140**

With $n = 8$: $\sum x = 410$, $\sum y = 1{,}095$, $\sum x^2 = 23{,}406$, $\sum y^2 = 151{,}811$, $\sum xy = 58{,}140$.

Step 1 — The three building blocks:

$$n\sum xy - \sum x \sum y = 8(58{,}140) - (410)(1{,}095) = 16{,}170$$

$$n\sum x^2 - \left(\sum x\right)^2 = 8(23{,}406) - 410^2 = 19{,}148$$

$$n\sum y^2 - \left(\sum y\right)^2 = 8(151{,}811) - 1{,}095^2 = 15{,}463$$

(a) Pearson's r:

$$r = \frac{16{,}170}{\sqrt{19{,}148 \times 15{,}463}} = \frac{16{,}170}{17{,}207.14} = \mathbf{0.940}$$

This indicates a **very strong positive** linear relationship between age and systolic blood pressure. *(Real clinical data are rarely this tidy — with only 8 adults, a correlation this high is unusual.)*

(b) Test of r:

$H_0 : \rho = 0$ (no linear relationship between age and SBP in the population)
$H_1 : \rho \neq 0$ (there is a linear relationship); $\alpha = 0.05$

$$t_{cal} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.9397\sqrt{8-2}}{\sqrt{1-0.9397^2}} = \frac{2.3018}{0.3420} = \mathbf{6.73}$$

$df = n - 2 = 6$, and from the t-table $t_{0.975,\,6} = 2.447$.

Since $|t_{cal}| = 6.73 > 2.447$, we **reject** H_0. SPSS would report Sig. (2-tailed) = 0.001, which is < 0.05 — the same decision.

Conclusion: There is a significant positive linear relationship between age and systolic blood pressure ($r = 0.94$, $df = 6$, $p = 0.001$): older adults tend to have higher SBP.

Simple Linear Regression

The Regression Equation

Simple linear regression models the relationship between X and Y as a straight line:

$$\hat{Y} = b_0 + b_1 X$$

Where:

Table 1: Symbols in the simple linear regression equation

Symbol	Name	Also called	Meaning	In Examples 1 and 2
Y	Outcome variable	**Dependent variable**, response variable	The variable we want to explain or predict; drawn on the vertical (y) axis	Systolic blood pressure (mmHg)
X	Predictor variable	**Independent variable**, explanatory variable	The variable we use to explain or predict Y; drawn on the horizontal (x) axis	Age (years)
$\hat{Y}$	Predicted value	Fitted value	The value of Y predicted by the line for a given X	Predicted SBP for a given age
b_0	Intercept	Constant	The predicted value of Y when $X = 0$	Predicted SBP at age 0 (not meaningful)
b_1	Slope	Regression coefficient	The change in $\hat{Y}$ for each one-unit increase in X	Change in predicted SBP per extra year of age

Which Variable Is Y and Which Is X?

Ask: *"Which variable do we want to predict?"* That one is the **dependent variable** Y, because its value is thought to **depend on** the other. The **independent variable** X usually comes first in time, is the possible cause, or is easier to measure. For example,

we use age (X) to predict blood pressure (Y), not the other way round.
For Pearson's r the choice does not matter (r is the same either way), but for regression it does: swapping X and Y gives a different line.
In SPSS (**Analyze** → **Regression** → **Linear**), put Y in the **Dependent** box and X in the **Independent(s)** box.

The line above is calculated from a **sample**. It estimates the true line in the **population**:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

where β_0 and β_1 (Greek "beta") are the population intercept and slope — **parameters** — and ε is random error. The sample values b_0 and b_1 are the **statistics** we use to estimate them, just as $\bar{x}$ estimates μ (Chapter 1).

For each patient, the **residual** e_i is the difference between the observed value and the value predicted by the sample line:

$$e_i = Y_i - \hat{Y}_i$$

Estimating the Coefficients

The values of b_0 and b_1 are chosen to minimize the **sum of squared residuals** — this method is called **Ordinary Least Squares (OLS)**:

$$\text{Minimize: } \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n} e_i^2$$

The OLS formulas use the **same column totals** as r ($\sum x$, $\sum y$, $\sum x^2$, $\sum xy$):

Formulas for b_1 and b_0 (Use These by Hand)

$$b_1 = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2}, \qquad b_0 = \bar{y} - b_1 \bar{x}$$

The numerator is the **same** as the numerator of r, and the denominator is the **first bracket** in the denominator of r — so if you have already calculated r, you can reuse both numbers. Keep at least **4 decimal places** in b_1 before calculating b_0.

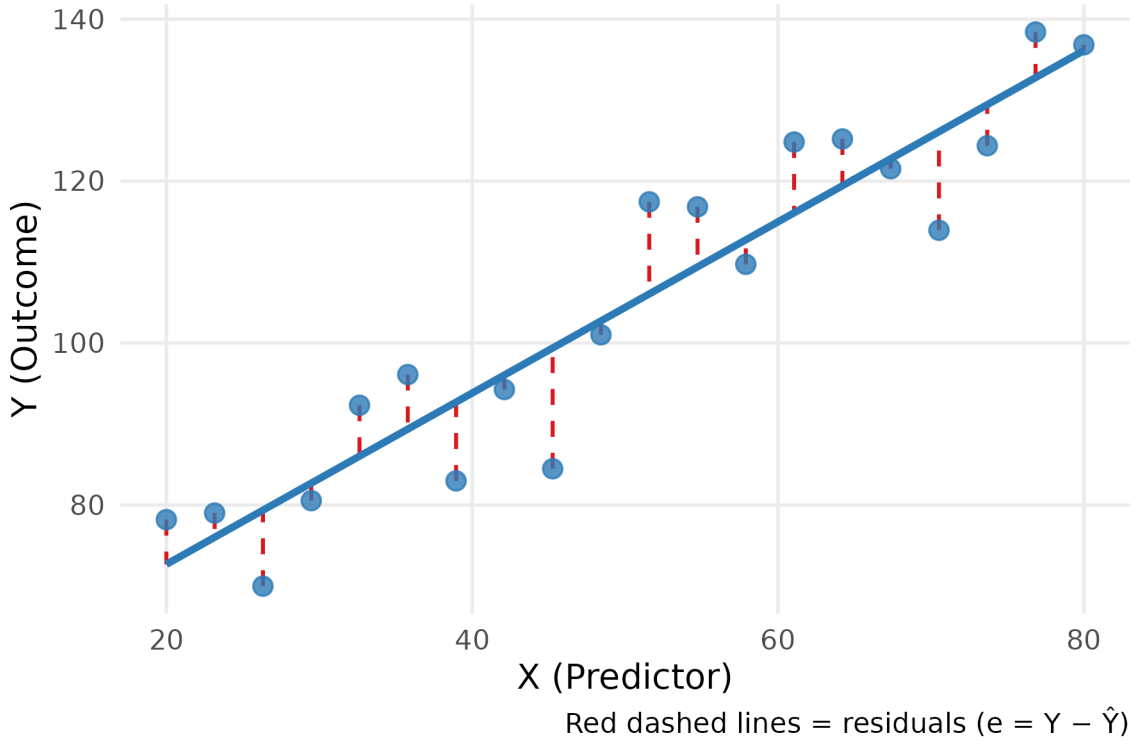


Figure 1: Least squares regression line minimizes the sum of squared residuals (vertical red lines)

The Coefficient of Determination (R^2)

R^2 (R-squared) measures how well the regression line fits the data — specifically, the **proportion of the total variation in Y that is explained by X**:

$$R^2 = r^2 \quad \text{(in simple linear regression)}$$

$$R^2 = \frac{\text{Variation explained by the model}}{\text{Total variation in } Y}$$

- $R^2 = 1$: the model explains 100% of the variation in Y (perfect fit)
- $R^2 = 0$: the model explains none of the variation in Y
- $R^2 = 0.65$ means: *"65% of the variation in Y is explained by X"*

Example 2: Fitting a Regression Line

Use the age and SBP data of the 8 adults in **Example 1** (and its column totals).

(a) Find the regression equation $\hat{Y} = b_0 + b_1 X$.

(b) Predict the SBP of a 50-year-old adult.

(c) Calculate and interpret R^2.

Solution

From Example 1: $n\sum xy - \sum x \sum y = 16{,}170$ and $n\sum x^2 - \left(\sum x\right)^2 = 19{,}148$.
(a) Regression coefficients (reusing the numbers from Example 1):

$$b_1 = \frac{16{,}170}{19{,}148} = 0.8445$$

$$\bar{x} = \frac{410}{8} = 51.25, \qquad \bar{y} = \frac{1{,}095}{8} = 136.875$$

$$b_0 = \bar{y} - b_1 \bar{x} = 136.875 - 0.8445 \times 51.25 = 93.59$$

$$\boxed{\hat{Y} = 93.59 + 0.8445X}$$

(b) Prediction for age = 50:

$$\hat{Y} = 93.59 + 0.8445 \times 50 = 135.8 \text{ mmHg}$$

(c) R^2:

$$R^2 = r^2 = (0.9397)^2 = 0.883$$

(Use r with 4 decimal places here; squaring the rounded 0.940 would give 0.884.)
88.3% of the variation in systolic blood pressure is explained by age. The remaining 11.7% is due to other factors not included in this model.

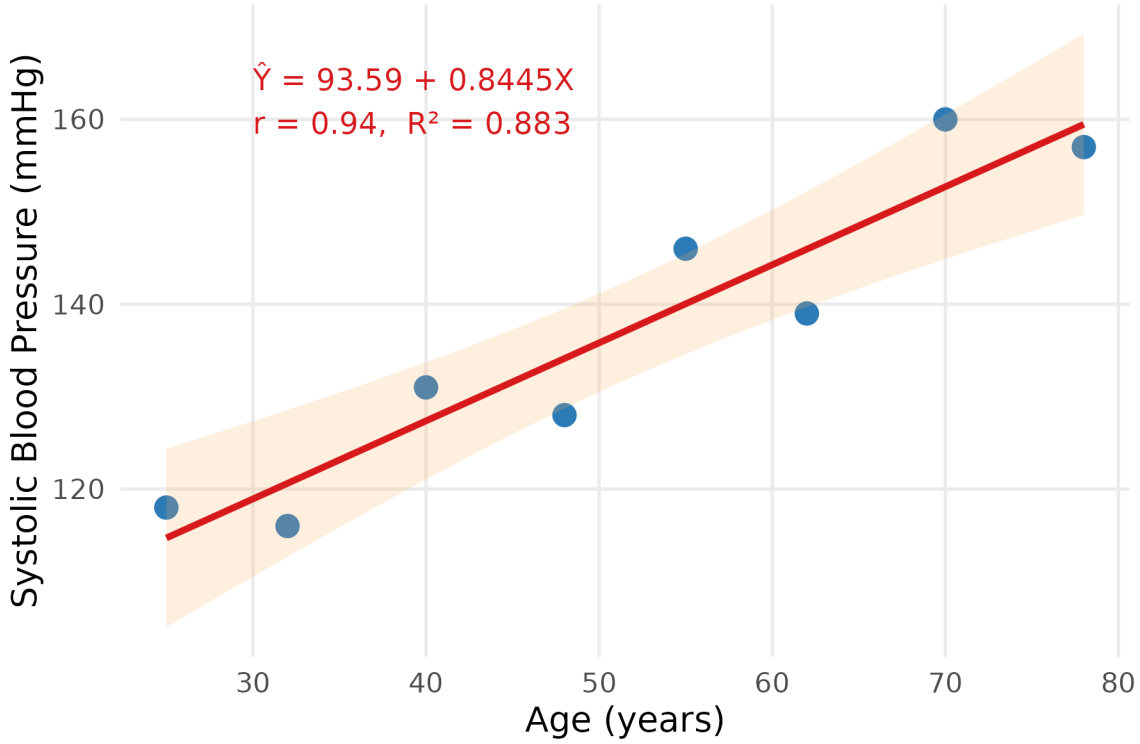


Figure 2: Scatter plot with fitted regression line: Age vs. Systolic Blood Pressure

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Testing the Significance of the Regression

Before using the regression equation for interpretation or prediction, we should test whether the population slope β_1 is different from zero. If $\beta_1 = 0$, then X provides no information about Y — the line is flat.

Hypotheses:

$$H_0 : \beta_1 = 0 \quad \text{(X is not a useful predictor of Y)}$$

$$H_1 : \beta_1 \neq 0 \quad \text{(X is a useful predictor of Y)}$$

The test statistic is:

$$t = \frac{b_1}{SE_{b_1}}$$

which follows a t-distribution with $df = n - 2$.

SPSS reports this p-value automatically in the **Coefficients table** of the regression output — read the row of the **predictor**, not the Constant row. In simple linear regression this test gives exactly the same p-value as the test of Pearson's r above: both ask whether there is a linear relationship between X and Y.

Assumptions of Linear Regression

The regression model is valid only when the following assumptions are met:

Table 2: Assumptions of Simple Linear Regression

Assumption	Meaning	How to Check
Linearity	The relationship between X and Y is linear (a straight line is appropriate)	Scatter plot of X vs. Y; residual plot
Independence	Observations are independent of each other	Study design - satisfied by random sampling
Homoscedasticity	The spread of residuals is roughly constant across all values of X	Plot of residuals vs. fitted values - no funnel shape
Normality of residuals	Residuals are approximately normally distributed	Histogram or Q-Q plot of residuals

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Reading SPSS Regression Output

A researcher studies whether **waist circumference (cm)** (X) predicts **fasting blood glucose (mg/dL)** (Y) in **40 adults** (**Analyze** → **Regression** → **Linear**). SPSS produces three key tables. Here is how to read them:

Table 3: SPSS Output: Model Summary

R	R2	Adjusted R2	Std. Error of the Estimate
0.821	0.674	0.666	8.458

R = 0.821 (very strong positive correlation); **R² = 0.674** — 67.4% of the variation in fasting glucose is explained by waist circumference. *(The Std. Error of the Estimate, 8.458 mg/dL, is the typical size of a residual; it is not needed in this course.)*

Table 4: SPSS Output: ANOVA Table

	Sum of Squares	df	Mean Square	F	Sig.
Regression	**5621.4**	**1**	**5621.4**	**78.62**	**< 0.001**
Residual	2718.6	38	71.5		
Total	8340.0	39			

The **ANOVA table** splits the total variation in Y (Total Sum of Squares) into the part explained by the line (Regression) and the part left over (Residual). R^2 = Regression SS / Total SS = 5621.4 / 8340.0 = 0.674.

The **F** statistic tests $H_0 : \beta_1 = 0$, the same hypothesis as the slope test. In simple linear regression, $F = t^2$ for the slope ($8.87^2 \approx 78.7$, the small difference is rounding), so the two tests always give the same p-value.

F = 78.62, p < 0.001 — the regression model is statistically significant.

Table 5: SPSS Output: Coefficients Table

						95% Confidence Interval for B	
	B	Std. Error	Beta	t	Sig.	Lower	Upper
Constant	32.15	6.24	-	5.15	< 0.001	19.52	44.78
Waist circumference	**0.87**	**0.10**	**0.821**	**8.87**	**< 0.001**	**0.67**	**1.07**

Regression equation: $\widehat{\text{Glucose}} = 32.15 + 0.87 \times \text{Waist}$

The slope **B = 0.87** is statistically significant ($p < 0.001$, read from the *Waist circumference* row): for each 1 cm increase in waist circumference, fasting glucose increases by **0.87 mg/dL** on average. The **Beta (standardized)** value equals r in simple regression and is not needed in this course.

The "95% Confidence Interval for B" Columns

When you tick *Confidence intervals* in SPSS, the Coefficients table gives a 95% CI for each B: $B \pm t_{0.975,\, n-2} \times$ Std. Error (here $t_{0.975,\, 38} = 2.024$).

- **Slope row:** we are 95% confident that each extra 1 cm of waist circumference

raises mean fasting glucose by between 0.67 and 1.07 mg/dL. The CI **does not include 0**, which agrees with Sig. < 0.05: the predictor is significant. (If the CI included 0, the slope would not be significant at the 0.05 level.)

- **Constant row:** a CI for the mean of Y when $X = 0$. Interpret it only if $X = 0$ is meaningful and close to the data. A waist of 0 cm is impossible, so here the Constant and its CI have no clinical meaning — predicting at $X = 0$ would be far outside the range of the data.

Example 3: Interpreting SPSS Output

A researcher uses SPSS to examine the relationship between **BMI (kg/m²)** and **fasting blood glucose (mg/dL)** in 60 adults. The following output is obtained:

Model Summary:

R	R^2	Adjusted R^2	Std. Error
0.612	0.375	0.364	14.23

Coefficients:

	B	Std. Error	t	Sig.
Constant	48.32	11.45	4.22	< 0.001
BMI	2.91	0.46	6.33	< 0.001

Interpretation

Correlation: $r = 0.612$ — a **strong positive** relationship between BMI and fasting blood glucose (0.60–0.79 in the guideline table).
Regression equation:

$$\hat{Y} = 48.32 + 2.91 \times \text{BMI}$$

Slope ($b_1 = 2.91$): For each 1 kg/m² increase in BMI, fasting blood glucose increases by **2.91 mg/dL** on average.
Intercept ($b_0 = 48.32$): The predicted fasting glucose when BMI = 0 is 48.32 mg/dL. *(Note: BMI of 0 is not a meaningful value in practice — the intercept is simply a mathematical anchor for the line.)*
$R^2 = 0.375$**:** BMI explains **37.5%** of the variation in fasting blood glucose. The remaining 62.5% is due to other factors (diet, genetics, physical activity, etc.).
Significance ($p < 0.001$): The slope is highly significant — BMI is a statistically

significant predictor of fasting blood glucose.
Prediction example: A person with BMI = 28 kg/m^2:

$$\hat{Y} = 48.32 + 2.91 \times 28 = 48.32 + 81.48 = 129.8 \text{ mg/dL}$$

Key Takeaways

Chapter 6 Summary

1. **Statistical modeling** describes the mathematical relationship between an outcome variable Y and one or more predictor variables X.
2. **Pearson's** r measures the strength and direction of a linear relationship between two continuous variables ($-1 \leq r \leq +1$). Its significance is tested with $H_0 : \rho = 0$ (SPSS: Sig. (2-tailed) in the Correlations table).
3. **Simple linear regression** fits the equation $\hat{Y} = b_0 + b_1 X$ using the least squares method, which minimizes the sum of squared residuals.
4. The **slope** b_1 tells us how much Y changes for each one-unit increase in X; the **intercept** b_0 is the predicted value of Y when $X = 0$.
5. R^2 is the proportion of variation in Y explained by the model. A higher R^2 means a better fit.
6. Always test whether the slope is significantly different from zero ($H_0 : \beta_1 = 0$) before using the model for interpretation. In simple regression the slope t-test, the ANOVA F-test ($F = t^2$) and the test of r all give the same p-value.
7. The model is only valid when the four assumptions — **linearity, independence, homoscedasticity, and normality of residuals** — are reasonably satisfied.

Formula Summary

Table 1: Formulas in this chapter

Quantity	Formula	Notes
Pearson correlation	$r = \dfrac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}}$	$-1 \leq r \leq 1$; needs $\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$, $\sum xy$

Quantity	Formula	Notes
Test of r	$t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	$H_0 : \rho = 0$; $df = n - 2$
Slope	$b_1 = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$	Same numerator as r; change in $\hat{Y}$ per 1-unit increase in X
Intercept	$b_0 = \bar{y} - b_1\bar{x}$	Predicted Y when $X = 0$
Prediction	$\hat{Y} = b_0 + b_1 X$	Only within the range of the data
Residual	$e_i = Y_i - \hat{Y}_i$	OLS minimises $\sum e_i^2$
Coefficient of determination	$R^2 = r^2 = \dfrac{\text{Regression SS}}{\text{Total SS}}$	Proportion of variation in Y explained
Test of the slope	$t = \dfrac{b_1}{SE_{b_1}}$	$H_0 : \beta_1 = 0$; $df = n - 2$; $F = t^2$

Exercises

Exercise 1

A study records the **hours of sleep per night** (X) and **systolic blood pressure (mmHg)** (Y) for 6 adults:

Subject	Sleep (hrs)	SBP (mmHg)
1	5	145
2	6	138
3	7	130
4	7	125
5	8	120
6	9	115

(a) Draw a scatter plot and describe the relationship.
(b) Make a table with columns x, y, x^2, y^2 and xy, then use the computational formula to calculate Pearson's r. What does it indicate?
(c) Test at $\alpha = 0.05$ whether the correlation is significant.
(d) Find the regression equation $\hat{Y} = b_0 + b_1 X$.
(e) Predict the SBP of a person who sleeps 6.5 hours per night.
(f) Calculate R^2 and interpret it.

Exercise 2

A researcher uses SPSS to study the relationship between a patient's **age (years)** and **total cholesterol (mg/dL)**. The SPSS output gives the following results:
Model Summary: R = 0.548, R^2 = 0.300
Coefficients:

	B	Std. Error	t	Sig.
Constant	120.4	18.6	6.47	< 0.001
Age	1.85	0.31	5.97	< 0.001

(a) Write the regression equation.
(b) Interpret the slope in the context of this study.
(c) What proportion of the variation in cholesterol is explained by age? Is this a strong model?
(d) Predict the total cholesterol of a 55-year-old patient.
(e) Is the predictor (age) statistically significant at $\alpha = 0.05$? How do you know?

Exercise 3 — Reflection Questions

Answer the following in 2–3 sentences each:
(a) A researcher finds that the correlation between ice cream sales and drowning deaths is $r = 0.85$. Does this mean eating ice cream causes drowning? Explain using the concept of correlation vs. causation.
(b) In a regression model predicting patient recovery time from age ($R^2 = 0.22$), a colleague says *"this model is useless because it only explains 22% of the variation."* Do you agree? What additional information would help evaluate this claim?
(c) Why is it generally not appropriate to use a regression equation to predict Y for values of X that are far outside the range of the original data?

End of Chapter 6

End of Course

Congratulations on completing all six chapters of Introduction to Biostatistics!
You have covered the full journey from understanding what data is, to summarizing it, to understanding distributions, making estimates, testing hypotheses, and finally modeling relationships between variables.
These tools form the foundation of evidence-based practice in health science.